

(need to change)Project 2 Linear Mixed Model

** **

March 15, 2026

1 Abstract

2 Introduction

A study was performed that measured heat flow for 24 participants, measuring their heat flow throughout four sessions with three activities performed per session, over two days at different temperatures. Measurements were taken every second during each session, but the data examined only reflects the mean heat flow per session. The goal of this analysis is to determine if there is significant inter- and intra-individual variability in heat flow measurements, and how the measurements from different sensors are related.

The longitudinal nature of the data suggests that a mixed-effects framework may be required. We approached the model-building process through three aspects. First, we used backwards selection and partial F-tests to identify significant predictors and determine the main fixed effects, and used partial F-tests to examine fixed effects for interaction terms as well. Secondly, we examined if there is inter-individual variability by exploring random intercepts for each individual, nested random intercepts for activity performed or specific sensor, and a random slope for day. Finally, we examined the underlying covariance structures of the data, testing whether there is temporal autocorrelation via an autoregressive structure (AR(1)), and using an ARH(1) structure to examine if there are heterogeneous residual variances across different sensors.

This report first discusses the Exploratory Data Analysis (EDA) performed, examining if there is missing data and performing an initial diagnostic check of an ordinary least squares model to determine if any transformations should be applied to the response variable. It then provides a theoretical overview of the methods used in the “Statistical Methods” section, discussing partial F-tests, hierarchical random effects, linear mixed models, and covariance matrices. It then discusses the results of the model-building process mentioned above, in the “Results” section. Finally, in the Discussion session, we used the final linear mixed model to assess the significant drivers of heat flow and the relationship between measurements.

```
> library(tidyverse) #installed tidyverse
> library(nlme)
> library(ggplot2)
> library(here)
> library(kableExtra)
> library(modelsummary)
> library(dplyr)
> library(performance)
>
> get_baseline_metrics <- function(orig, name) {
+   data.frame(
+     Model = name,
+     Df = attr(logLik(orig), "df"),
+     AIC = round(AIC(orig)),
+     BIC = round(BIC(orig)),
+     LogLik = round(as.numeric(logLik(orig))),
+   )
+ }
```

```

+   stringsAsFactors = FALSE
+ )
+ }
>
> get_anova_outputs <- function(orig, new, name) {
+   res <- anova(orig, new) #, type = "marginal" doesnt matter if pass 2 models; uses chi-squared
+   data.frame(
+     Model = name,
+     Df = attr(logLik(new), "df"),
+     AIC = round(res$AIC[2]),
+     BIC = round(res$BIC[2]),
+     LogLik = round(res$logLik[2]),
+     LRT_Stat = round(res$L.Ratio[2], 2),
+     p_value = res$p-value[2]
+   )
+ }

```

3 Exploratory Data Analysis

An exploratory data analysis was conducted, aiming to understand the distributional properties of the heat flow measurements and identify preliminary trends regarding inter-individual and intra-individual variability, as well as the relationships between the different sensors.

3.1 Dataset Overview

The dataset consists of measurements from 24 participants (12 female, 12 male) monitored over two separate days. The only environmental difference between Day 1 and Day 2 is the temperature: Day 1 is 24 °C while Day 2 is 18 °C. While the original experiment captured data every second, this analysis utilizes the mean heat flow per activity in each session monitored by 6 sensors to address the research goals. We notice that for day 2 session 2, only sitting is measured; while in day 1 session 2, both sitting and standing are measured.

98 out of 2736 observations have missing heat flow measurement and we directly dropped them.

All variables are categorical: **day** has 2 levels ("1" and "2"); **session** has 4 levels ("1", "2", "3" and "4"); **activity** has 3 levels ("sitting", "standing" and "walking"); **sex** has 2 levels ("F" and "M"); **HF_type**(sensor ID) has 6 levels ("HF_2", "HF_3", "HF_5", "HF_7", "HF_8", "HF_16").

3.2 Visual Evidence for Fixed Effects

To determine which variables influence heat flow, we examined the measurements across different experimental conditions. The exploratory plots indicate that heat flow is not constant and is influenced by every categorical variable in the study.

Figure 1 displays heat flow during the sitting activity, partitioned by Day, Session, Sensor, and Sex.

For Sensor Type, we can see there are distinct vertical offsets between panels (e.g., HF_16 vs. HF_1), showing that different body locations have different baseline heat flow. Comparing the top row (Day 1, 24°C) to the bottom row (Day 2, 18°C) reveals a clear shift in mean heat flow, confirming a strong environmental effect. The solid linear fit lines across the four sessions often show non-zero slopes, suggesting that heat flow changes over the course of the day even when activity is held constant. The red (Female) and blue (Male) lines are frequently separated, indicating that biological sex influences thermal regulation.

```

> # (Below is not needed anymore for checking linearity, we treat session as
> # factor)
> # make session to be continuous variable for now

```

```

> # spaghetti plot + faceting
> data_sitting <- data_clean %>%
+   filter(activity == "sitting")
>
> ggplot(data_sitting, aes(x = session_num, y = HF_value, color = sex)) +
+   geom_point(alpha = 0.1) +
+
+   # Flexible LOESS curves (Dashed lines, one for M and one for F)
+   geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linetype = "dashed", size = 0.5) +
+
+   # Straight Linear regression lines (Solid lines, one for M and one for F)
+   geom_smooth(method = "lm", formula = y ~ x, se = FALSE, linetype = "solid", size = 0.5) +
+
+   # Grid: Rows = Activity, Columns = Sensor
+   facet_grid(day ~ HF_type) +
+
+   # Manually set colors for clarity (e.g., Males = Blue, Females = Red)
+   scale_color_manual(values = c("M" = "blue", "F" = "red")) +
+
+   theme_minimal() +
+   labs(
+     title = "(Sitting) Heat Flow Over Time by Day, Session, Sensor and Sex",
+     subtitle = "Solid lines = Linear Fit (lm) | Dashed lines = Flexible Fit (loess)",
+     x = "Session (1 to 4)",
+     y = "Heat Flow (HF_value)",
+     color = "Sex"
+   )

```

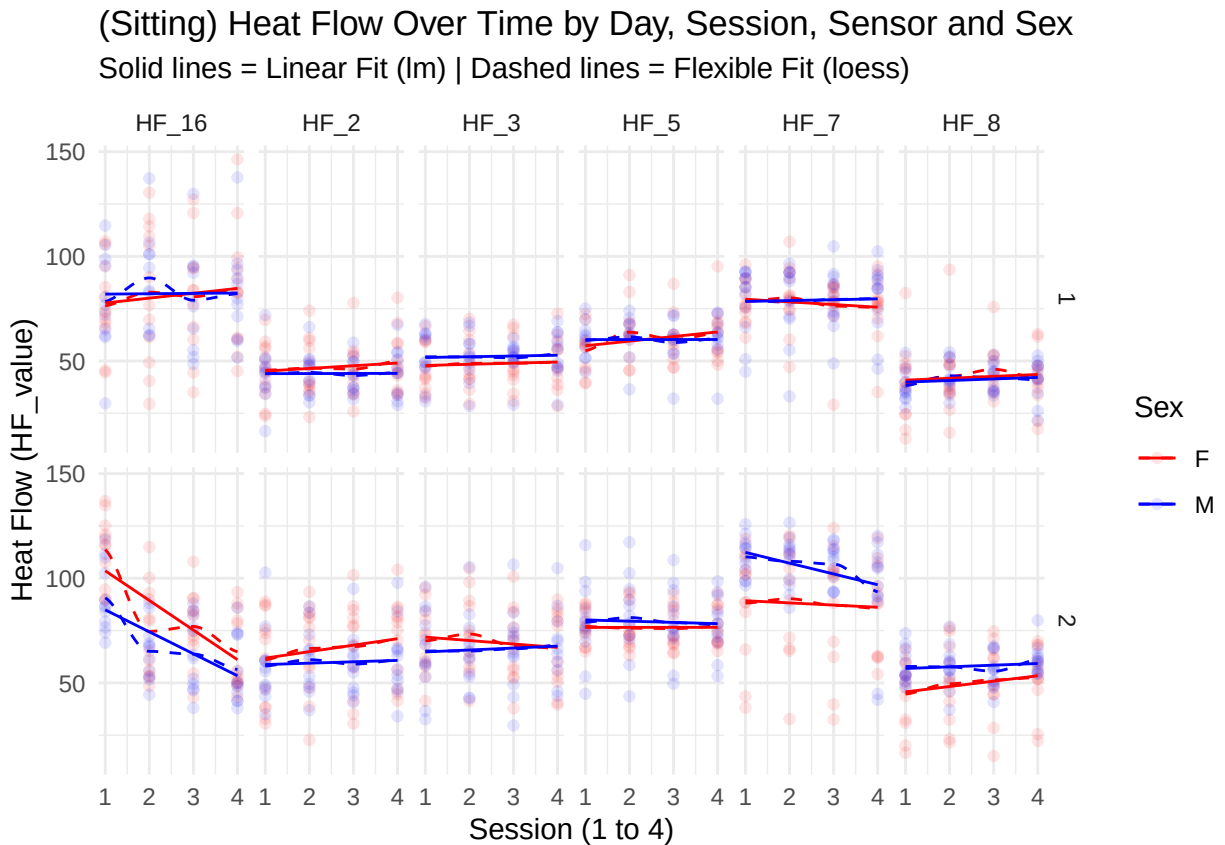


Figure 1: Sitting's Heat Flow Over Time by Day, Session, Sensor and Sex

Figure 2 focuses on Day 1 to highlight the impact of activity: Comparing the rows for sitting, standing, and walking reveals that heat flow varies with physical movement.

```
> data_day1 <- data_clean %>% filter(day == "1")
>
> ggplot(data_day1, aes(x = session_num, y = HF_value, color = sex)) +
+   geom_point(alpha = 0.1) +
+   geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linetype = "dashed", size = 0.5) +
+   # Straight Linear regression lines (Solid lines, one for M and one for F)
+   geom_smooth(method = "lm", formula = y ~ x, se = FALSE, linetype = "solid", size = 0.5) +
+   facet_grid(activity ~ HF_type) +
+   scale_color_manual(values = c("M" = "blue", "F" = "red")) +
+   theme_minimal() +
+   labs(
+     title = "(Day 1)Heat Flow Over Time by Activity, Sensor and Sex",
+     subtitle = "Solid lines = Linear Fit | Dashed = LOESS | Grid = Activity+Day vs Sensor",
+     x = "Session (1 to 4)",
+     y = "Heat Flow (HF_value)",
+     color = "Sex"
+   )
```

+)

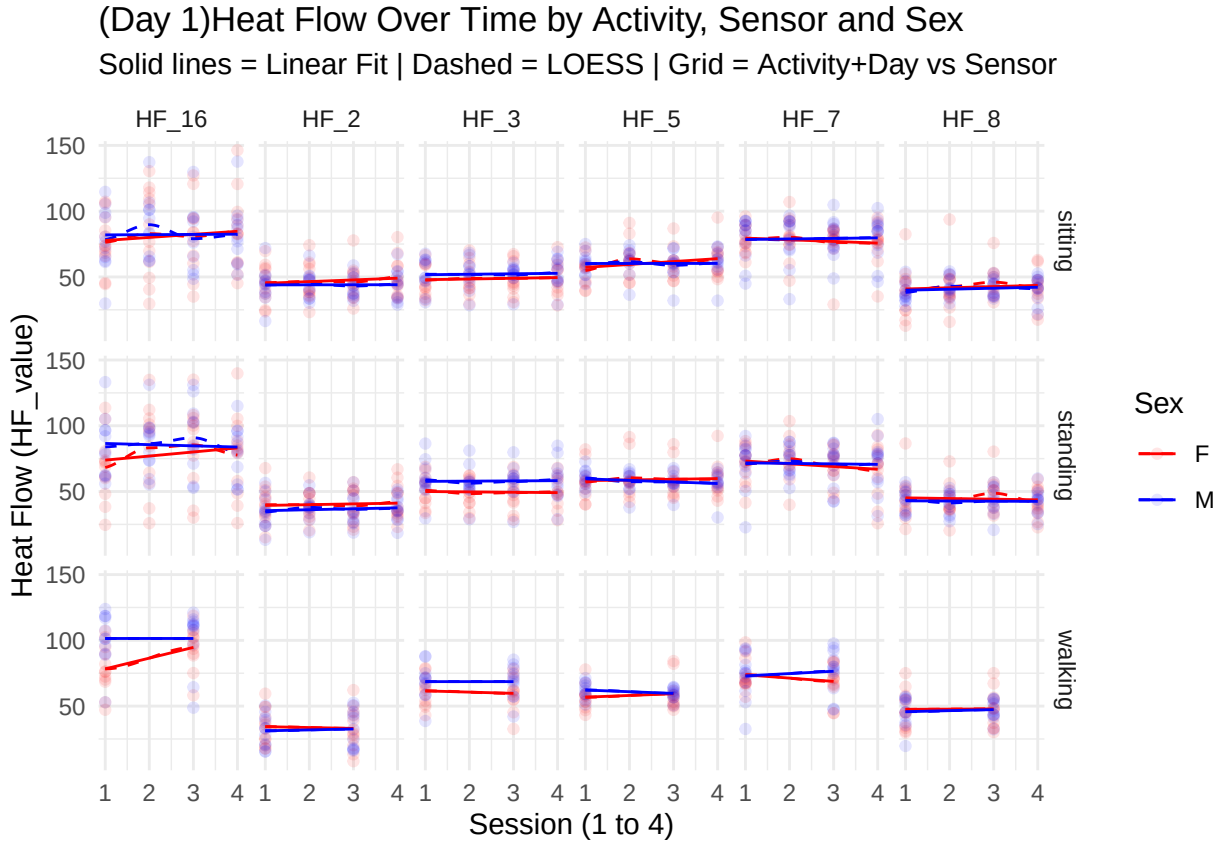


Figure 2: Day 1's Heat Flow Over Time by Activity, Sensor and Sex

Because the plots and physiological intuition suggest that every categorical variable exerts a fixed effect on the response, all predictors were carried forward into the modeling stage.

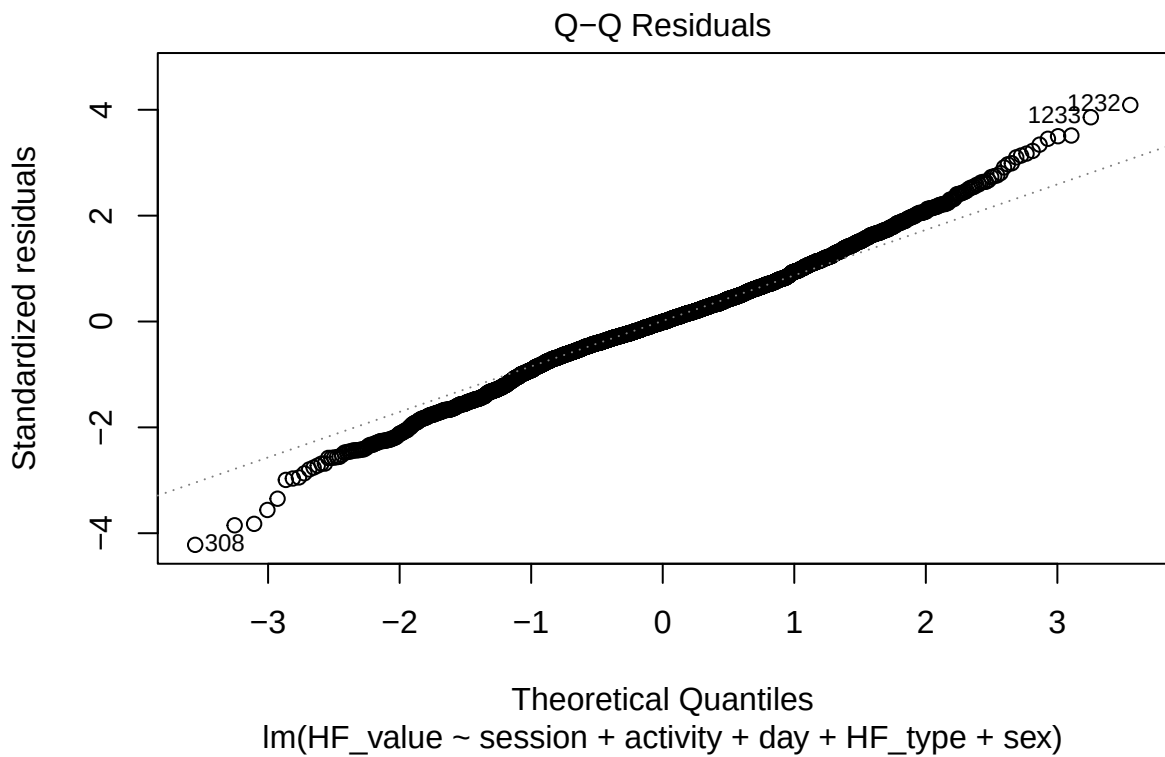
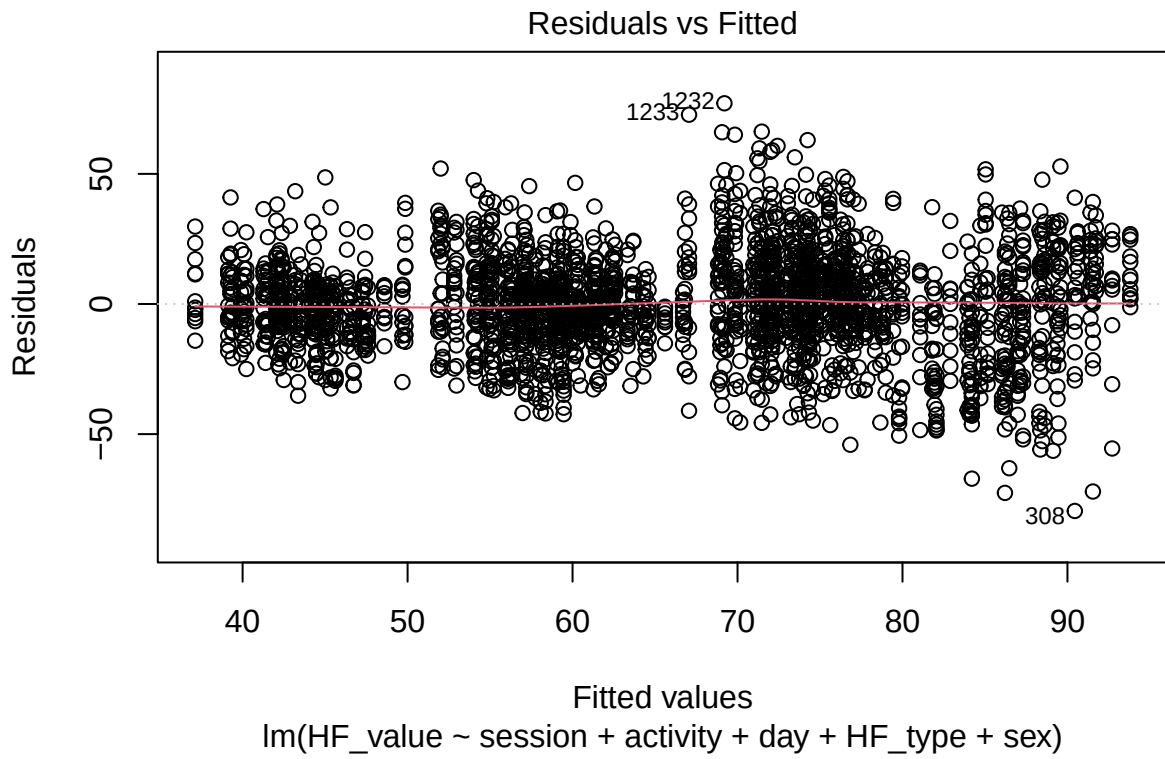
4 Statistical Description of the Methods

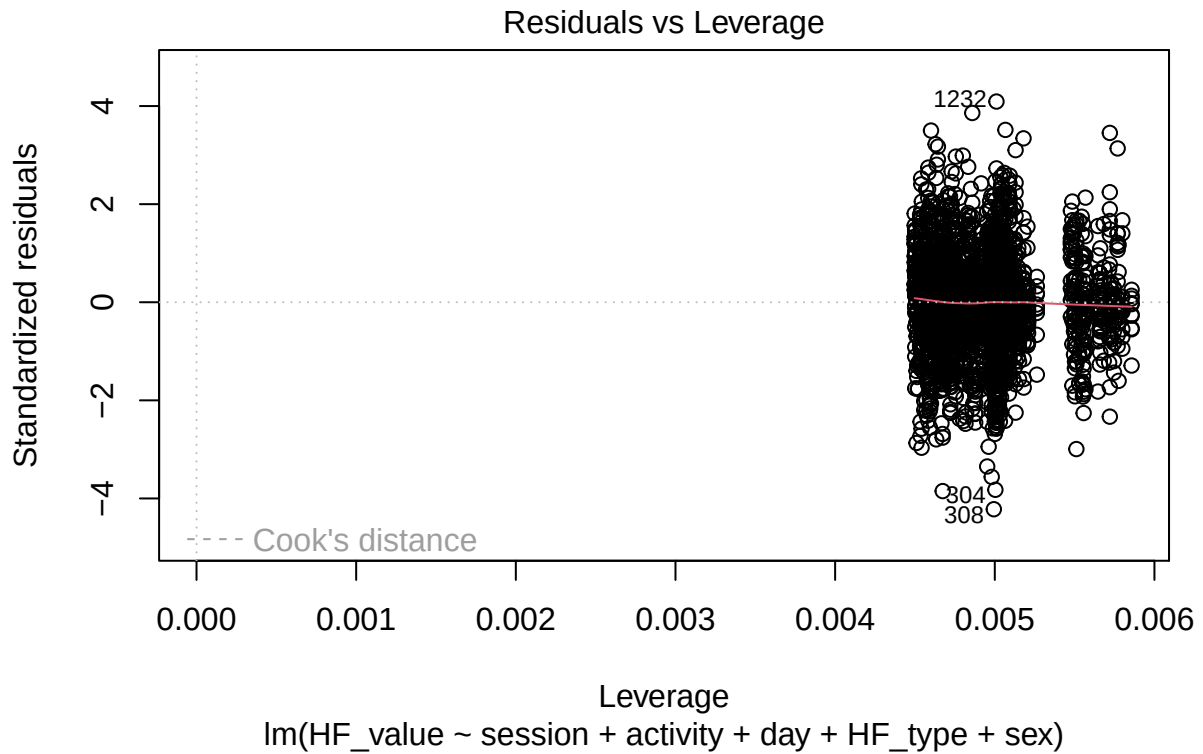
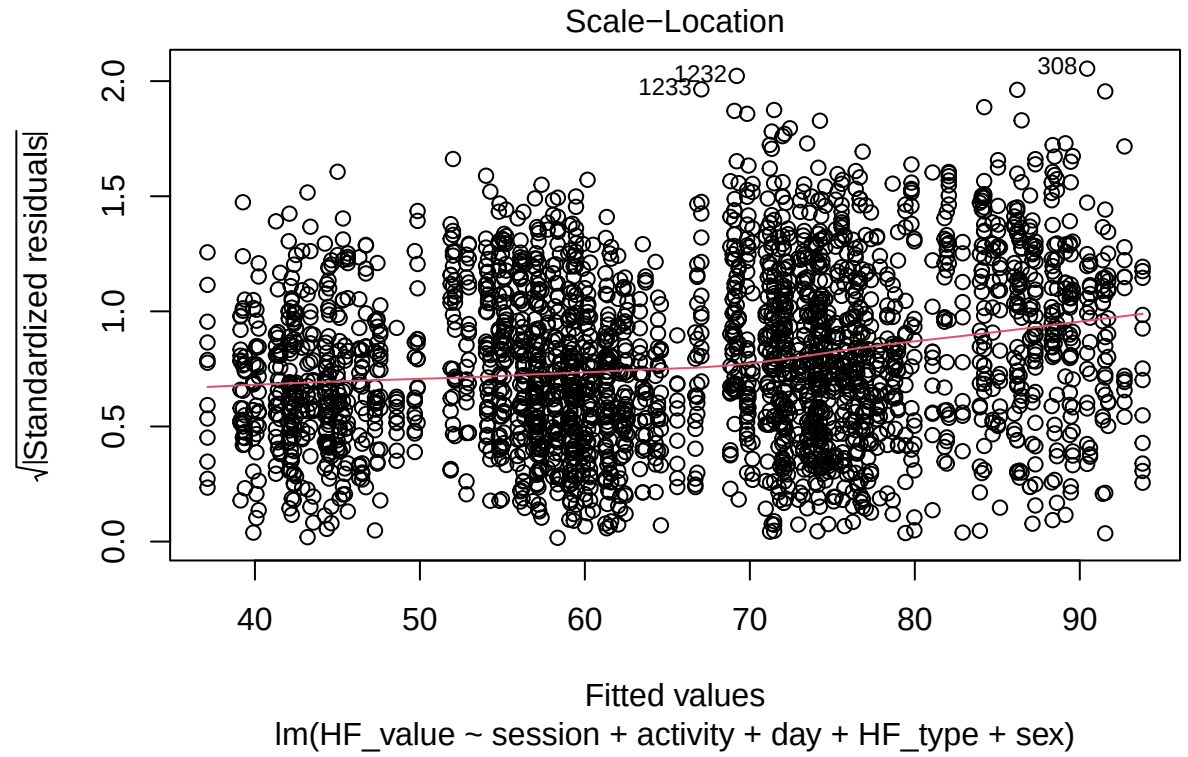
To analyze the factors influencing human heat flow while accounting for the complex hierarchical and temporal dependencies inherent to the experimental design, a Linear Mixed Model (LMM) was employed.

4.1 Log Transformation

An initial naive ordinary least squares (OLS) regression was fitted to identify the baseline relationships between the predictors and the response variable and justify the inclusion of all 5 variables. Diagnostic plots of this initial model revealed significant heteroscedasticity and heavy-tailed residuals. To satisfy the assumptions of linear modeling and stabilize the variance, a natural logarithm transformation was applied to the heat flow measurements ($Y = \log(\text{HF_value})$). Post-transformation diagnostics revealed that while the residuals still exhibit a slightly heavier lower tail, the transformation successfully reduced severe heteroscedasticity present in the untransformed data, which is the higher priority for valid statistical inference. Subsequent analyses were conducted on this log scale.

```
> plot(naive_lm)
```





4.2 F Tests for Fixed Effect Selection

To determine the optimal fixed-effects structure, a systematic, multi-step model-building approach was employed. First, partial F-tests were utilized to evaluate the significance of predictors in a main-effects-only model.

For a given categorical predictor or interaction term with r degrees of freedom (corresponding to $k - 1$ coefficients for a variable with k levels), the partial F-test evaluates whether the inclusion of the term provides a statistically significant improvement to the model's goodness-of-fit. The null hypothesis posits that all r coefficients associated with the specific term are simultaneously zero:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_r = 0$$

To test this, the F-statistic is calculated by comparing the Residual Sum of Squares (RSS) of a reduced model (excluding the term in question) against a full model (including the term):

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/r}{RSS_{\text{full}}/(n - p)}$$

(Where n represents the total number of observations, and p represents the total number of parameters estimated in the full model. Note that r is equivalent to $p_{\text{full}} - p_{\text{reduced}}$).

Under the null hypothesis, this test statistic follows an F-distribution with r and $n - p$ degrees of freedom ($F_{r, n-p}$). The mathematical validity of this test assumes that the model's residual errors are independent, identically distributed, and follow a normal distribution, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

Next, to identify relevant interaction terms, a preliminary marginal screening of ten theoretically plausible two-way interactions was conducted. To test whether the addition of each interaction term significantly improved the model (Null Hypothesis (H_0): The coefficient for the interaction term is exactly zero, $\beta_{\text{interaction}} = 0$), individual partial F-tests were performed.

Because conducting multiple simultaneous hypothesis tests artificially inflates the probability of Type I errors (false positives), a Bonferroni correction was applied to control the family-wise error rate (FWER). Based on a standard family-wise alpha level of 0.05 distributed across 10 tests, the adjusted significance threshold was established at $p < 0.005$.

To finalize the fixed-effects structure, a marginal backward elimination procedure (using partial F-test) was conducted for the resulting model. Furthermore, to adhere to the principle of marginality, any main effect involved in a significant two-way interaction was retained in the model, regardless of its independent main-effect significance.

4.3 Hierarchical Random-Effects Structure

The experimental design involved repeated measurements from the same 24 participants across multiple days, sessions, and physical activities. An assumption of standard linear regression is that all observations are independent. However, because multiple heat flow measurements were taken from the same individuals, unobserved subject-specific physiological traits (such as baseline metabolic rate or core temperature) were expected to cause intra-subject correlation.

To appropriately account for this unobserved heterogeneity and resolve the violation of independence, a Linear Mixed Model (LMM) was employed.

Mathematically, an LMM extends the general linear model by partitioning the total variance into fixed and random components. In matrix notation, the general model is defined as:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{b} + \epsilon$$

Where:

$\mathbf{y}$ is the vector of the response variable ($\log(\text{HF_value})$).

$\mathbf{X}$ is the design matrix for the fixed effects (the controlled experimental conditions such as Day, Session, and Activity).

β is the vector of fixed-effect coefficients, representing the population-average responses.

$\mathbf{Z}$ is the design matrix for the random effects. This matrix can contain both indicator variables for grouping factors (yielding random intercepts) and continuous/categorical covariates (yielding random slopes).

$\mathbf{b}$ is the vector of random effects. In this study, it captures both a subject’s unique baseline shift (a random intercept) and their specific physiological trajectory in response to a changing experimental condition (a random slope). It is assumed to be normally distributed with mean zero and covariance matrix $\mathbf{G}$, such that $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{G})$.

ϵ is the vector of residual errors, assumed to be normally distributed with mean zero and covariance matrix $\mathbf{R}$, such that $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$.

By introducing the random effects vector $\mathbf{b}$, the LMM defines the marginal covariance matrix of the response $\mathbf{y}$ as:

$$Var(\mathbf{y}) = \mathbf{ZGZ}^T + \mathbf{R}$$

Unlike standard Ordinary Least Squares (OLS) regression, which assumes a strictly diagonal covariance matrix where all observations are completely independent, this $\mathbf{ZGZ}^T$ structure mathematically permits non-zero covariances between observations belonging to the same participant. Furthermore, by allowing $\mathbf{G}$ to estimate both random intercepts and random slopes, the model accounts for the fact that participants not only start at different thermal baselines, but also adapt to the environmental temperature changes at entirely different physiological rates.

4.4 Residual Variance and Temporal Correlation

In standard linear mixed models, the residual errors (denoted as ϵ_{ijk} for participant i , sensor type j , and sequential time step k) are conventionally assumed to be conditionally independent and identically distributed with a constant variance σ^2 . However, given the longitudinal nature of this experimental design, the assumption of residual independence may be overly restrictive. Physiologically, heat flow is a continuous, dynamic process; therefore, it is hypothesized that short-term, unexplained biological variations exhibit temporal autocorrelation. Specifically, an unexpected fluctuation in a participant’s heat flow during one physical activity is likely to influence their physiological state in the subsequent activities.

To mathematically account for this sequential effect, a first-order Autoregressive AR(1) correlation structure was considered. A discrete ordinal time step (1 through 10) was constructed to represent the strict chronological sequence of the physical activities. This temporal memory was explicitly nested at the daily level (person/HF_type/day). This boundary specification ensures that residual autocorrelation decays sequentially across all activities within a single day, but assumes complete mathematical independence across the overnight gap between Day 1 and Day 2.

Let ρ be the autoregressive parameter, and $|k - k'|$ be the chronological time lag between two measurement instances. Under AR(1) assumption, the covariance between any two residuals from the same sensor on the same person is defined as:

$$Cov(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

Where the indicator function $I(\cdot)$ returns 1 if the measurements occurred on the same day, and 0 if they occurred on different days, breaking the temporal correlation overnight.

While the AR(1) specification successfully captures temporal memory, it still imposes a strict assumption of homoscedasticity, forcing a single, universal residual variance (σ^2) across all observations. Physiologically, different sensor placements (e.g., wrist versus chest) experience varying degrees of ambient noise interference and physical contact stability.

To resolve this limitation, the model's residual structure was upgraded to an Autoregressive Heteroscedastic ARH(1) formulation. This structure retains the autoregressive sequential decay (ρ) but relaxes the constant variance assumption by calculating a unique, independent residual variance (σ_j^2) for each specific sensor type j .

The final residual errors are distributed as a vector $\epsilon_{ij} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{ij})$, where the elements of the block-diagonal residual covariance matrix $\mathbf{R}_{ij}$ are formalized as:

$$\text{Cov}(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma_j^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

This final $\mathbf{R}$ matrix allows the baseline magnitude of the physiological noise to scale up or down depending on the specific physical sensor, while perfectly preserving the sequential decay of that heat flow noise across the entire experimental day.

4.5 Likelihood Ratio Tests for Model Selection

Likelihood Ratio Tests (LRT) were employed to compare nested models and justify the inclusion of specific random effects, correlation structures, and fixed covariates. The LRT statistic evaluates whether additional parameters significantly improve model fit, defined as:

$$LRT = -2 \ln \left(\frac{L_{reduced}}{L_{full}} \right)$$

Under the null hypothesis (H_0 : the extra parameters in the complex model are equal to zero, and thus add no predictive value), this statistic asymptotically follows a Chi-square (χ^2) distribution, with degrees of freedom equal to the difference in the number of freely estimated parameters between the two models.

Models were primarily fitted using Restricted Maximum Likelihood (REML) to obtain unbiased estimates of the variance and covariance components. However, because REML likelihoods are dependent on the fixed-effects design matrix, they cannot be used to compare models with different fixed effects. Therefore, the following testing protocol was utilized: LRTs evaluating random effects and residual structures were conducted using REML (holding fixed effects constant), whereas LRTs used to prune fixed-effect covariates were conducted after temporarily refitting the competing models with standard Maximum Likelihood (ML).

It is noted that testing whether a random-effect variance is exactly zero places the parameter on its mathematical boundary, making standard unadjusted p-values conservative. Because the LRT p-values for all retained variance components in this study were exceptionally small ($p < 0.0001$), manually adjusting them (dividing by 2) did not alter any statistical conclusions.

Alongside LRTs, the Akaike (AIC) and Bayesian (BIC) Information Criteria were continuously evaluated to optimize model fit while penalizing unnecessary mathematical complexity.

5 Results

```
> par(mfrow = c(2, 2), mar = c(4, 4, 3, 2), oma = c(5, 0, 3, 0))
> #2 rows 2 columns
> #oma = outer margins to have space for caption
>
> #NAIVE (row 1)
> # Plot 2: Normal Q-Q
> plot(naive_lm, which = 2, main = "Model 0 (Raw): Normality (QQ)", sub = "", caption = "") #sub means
> # Plot 3: residuals vs fitted
> plot(naive_lm, which = 1, main = "Model 0 (Raw): Residuals vs Fitted Values", sub = "", caption = "")
>
> #LOG (row 2)
```

```

> # Plot 2: Normal Q-Q
> plot(log_naive_lm, which = 2, main = "Model 1 (Log): Normality (QQ)", sub = "", caption = "")
> # Plot 3: residuals vs fitted
> plot(log_naive_lm, which = 1, main = "Model 1 (Log): Residuals vs Fitted Values", sub = "", caption =
>
>
> # caption_text<-
>
> # wrapped_caption <- paste(strwrap(caption_text, width = 150), collapse = "\n")
>
> mtext("Diagnostic Comparison of Raw & Log Linear Models",
+       side = 3, line = 0, outer = TRUE, cex = 1.2, font = 2)

```

Diagnostic Comparison of Raw & Log Linear Models

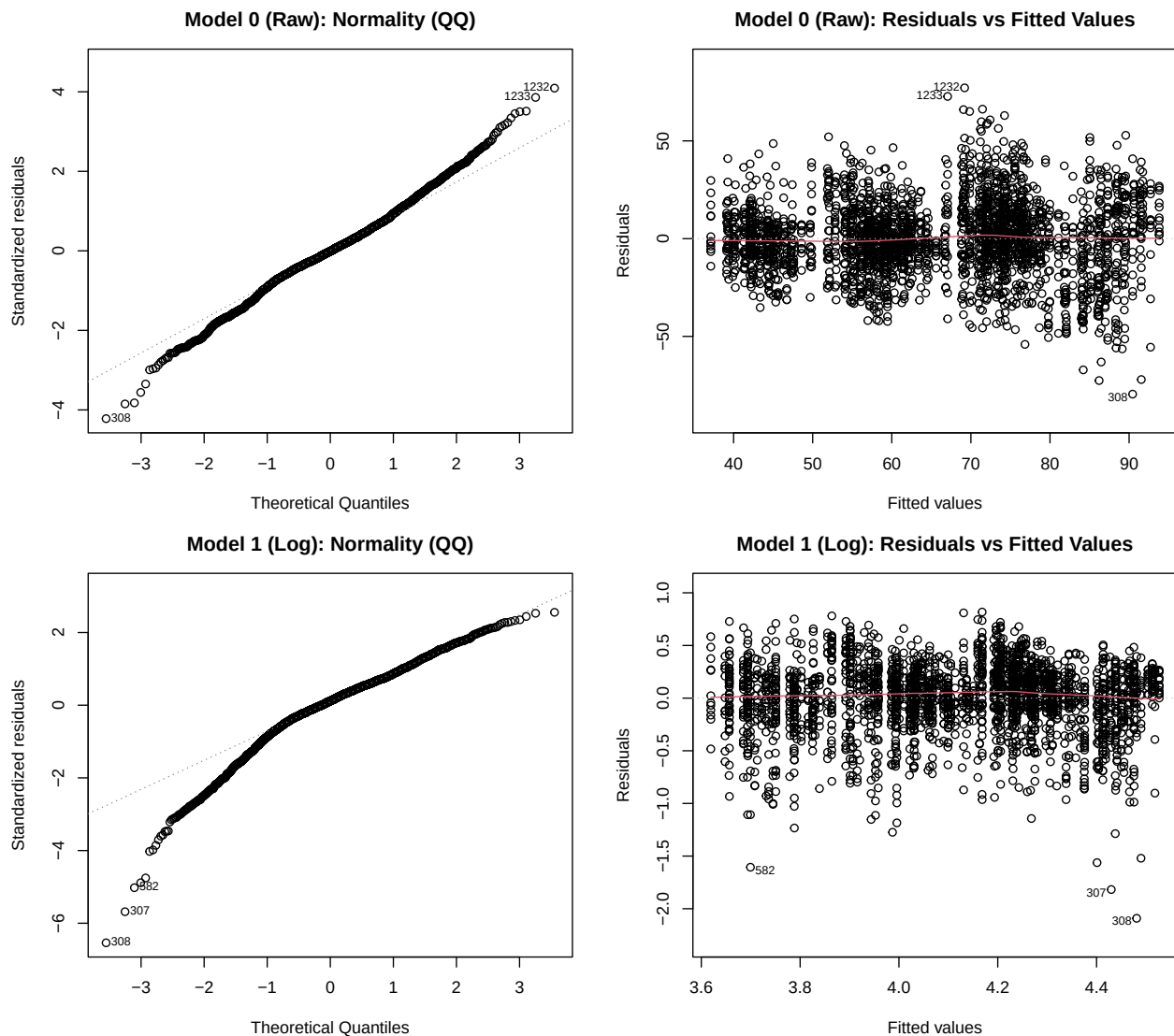


Figure 3: The QQ-plot of the Model 0 (the raw (untransformed) model) shows an approximately normal model with slightly heavy tails, but the Residuals vs Fitted Values plot shows heteroscedastic residuals, with variance increasing as fitted values increase. Model 1 (the log-transformed model) corrects the right heavy tail but the left tail deviates more, and the Scale-Location plot shows fairly homoscedastic residuals.

```
> # mtext(wrapped_caption,
> #       side = 1, line = 2, outer = TRUE, cex = 0.8)
```

As seen in Figure 3 applying a simple linear model ($\text{HF_value} \sim \text{session} + \text{activity} + \text{day} + \text{HF_type} + \text{sex}$) shows that the residuals are approximately normal (with heavy tails), but heteroscedastic. Transforming the response variable ($\log(\text{HF_value})$) resulted in homoscedasticity and corrected the heavy right tail, but introduced more left-skew in the residuals. We prioritized homoscedasticity and use the log-transformed model going forward, to ensure reliable parameter estimation and significance testing.

5.1 Fixed Effects

Initial partial F-tests on the main-effects model ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$) indicated that the session variable was independently non-significant and was flagged for initial removal. However, the subsequent preliminary screening of the ten two-way interactions isolated four interactions that successfully met the rigorous Bonferroni-adjusted threshold ($p < 0.005$): day & sensor, session & sensor, activity & sensor, and sex & sensor. Because the session & sensor interaction proved significant ($p = 0.002$), the main effect for session was added back into the model to preserve structural marginality.

A fixed-effects model was then constructed containing all main effects and these four interaction terms ($\log(\text{HF_value}) \sim (\text{activity} + \text{day} + \text{sensor} + \text{sex}) * \text{sensor}$). A marginal backward elimination procedure (using partial F-tests) was conducted on this model. The results demonstrated that all four tested two-way interactions with the sensor (HF_type) remained significant ($p < 0.005$ across all terms). Consequently, no terms were eliminated at this stage. This suggests that each sensor reacts differently to the day, session, physical activity, and biological sex of an individual.

5.2 Random Intercept

```
> library(nlme)
> m1 <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+         random = ~ 1 | person, #unique intercept for each person
+         data = data_clean,
+         method = "REML")
>
> #summary(m1)

> m0_gls_reml <- gls(log(HF_value) ~ (day + session + activity + sex) * HF_type,
+                   data = data_clean,
+                   method = "REML")
>
>
> summary(m1)
```

Linear mixed-effects model fit by REML

Data:	data_clean
AIC	BIC
1207.812	1500.783
	-553.9061

Random effects:

Formula:	~1 person
	(Intercept) Residual
StdDev:	0.08333832 0.2844007

Fixed effects:	$\log(\text{HF_value}) \sim (\text{day} + \text{session} + \text{activity} + \text{sex}) * \text{HF_type}$				
	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.479602	0.04418779	2568	101.37648	0.0000

day2	-0.165397	0.02753267	2568	-6.00732	0.0000
session2	-0.068984	0.04462140	2568	-1.54599	0.1222
session3	-0.067929	0.03417981	2568	-1.98740	0.0470
session4	-0.201175	0.03989567	2568	-5.04252	0.0000
activitystanding	-0.087103	0.03114911	2568	-2.79631	0.0052
activitywalking	0.038501	0.03934063	2568	0.97866	0.3278
sexM	0.004110	0.04357422	22	0.09433	0.9257
HF_typeHF_2	-0.711877	0.05217546	2568	-13.64390	0.0000
HF_typeHF_3	-0.641528	0.05277767	2568	-12.15529	0.0000
HF_typeHF_5	-0.410268	0.05298564	2568	-7.74300	0.0000
HF_typeHF_7	-0.156055	0.05196167	2568	-3.00328	0.0027
HF_typeHF_8	-0.892809	0.05235319	2568	-17.05358	0.0000
day2:HF_typeHF_2	0.492416	0.03875616	2568	12.70549	0.0000
day2:HF_typeHF_3	0.501060	0.03903910	2568	12.83484	0.0000
day2:HF_typeHF_5	0.427314	0.03924014	2568	10.88972	0.0000
day2:HF_typeHF_7	0.342664	0.03859369	2568	8.87875	0.0000
day2:HF_typeHF_8	0.479292	0.03886928	2568	12.33086	0.0000
session2:HF_typeHF_2	0.126035	0.06257996	2568	2.01399	0.0441
session3:HF_typeHF_2	0.101480	0.04812013	2568	2.10888	0.0351
session4:HF_typeHF_2	0.283330	0.05629551	2568	5.03290	0.0000
session2:HF_typeHF_3	0.046317	0.06300399	2568	0.73514	0.4623
session3:HF_typeHF_3	0.042632	0.04837859	2568	0.88121	0.3783
session4:HF_typeHF_3	0.184793	0.05677569	2568	3.25479	0.0011
session2:HF_typeHF_5	0.086032	0.06344480	2568	1.35602	0.1752
session3:HF_typeHF_5	0.061820	0.04864707	2568	1.27079	0.2039
session4:HF_typeHF_5	0.198626	0.05691482	2568	3.48988	0.0005
session2:HF_typeHF_7	0.067088	0.06246244	2568	1.07406	0.2829
session3:HF_typeHF_7	0.023360	0.04787113	2568	0.48799	0.6256
session4:HF_typeHF_7	0.090332	0.05608570	2568	1.61061	0.1074
session2:HF_typeHF_8	0.093123	0.06260596	2568	1.48745	0.1370
session3:HF_typeHF_8	0.128704	0.04834497	2568	2.66219	0.0078
session4:HF_typeHF_8	0.232480	0.05630772	2568	4.12874	0.0000
activitystanding:HF_typeHF_2	-0.091764	0.04379096	2568	-2.09549	0.0362
activitywalking:HF_typeHF_2	-0.377798	0.05545474	2568	-6.81272	0.0000
activitystanding:HF_typeHF_3	0.189686	0.04412383	2568	4.29895	0.0000
activitywalking:HF_typeHF_3	0.232009	0.05577278	2568	4.15989	0.0000
activitystanding:HF_typeHF_5	0.062717	0.04433521	2568	1.41462	0.1573
activitywalking:HF_typeHF_5	-0.038941	0.05607659	2568	-0.69442	0.4875
activitystanding:HF_typeHF_7	-0.043330	0.04362560	2568	-0.99323	0.3207
activitywalking:HF_typeHF_7	-0.120250	0.05517846	2568	-2.17930	0.0294
activitystanding:HF_typeHF_8	0.182402	0.04394940	2568	4.15027	0.0000
activitywalking:HF_typeHF_8	0.166013	0.05562698	2568	2.98440	0.0029
sexM:HF_typeHF_2	-0.074248	0.03833085	2568	-1.93702	0.0529
sexM:HF_typeHF_3	0.083189	0.03859034	2568	2.15569	0.0312
sexM:HF_typeHF_5	0.013351	0.03881501	2568	0.34396	0.7309
sexM:HF_typeHF_7	0.113521	0.03815055	2568	2.97562	0.0030
sexM:HF_typeHF_8	0.074246	0.03843794	2568	1.93159	0.0535
Correlation:					
	(Intr)	day2	sessn2	sessn3	sessn4
day2					
session2	-0.336				
session3	-0.453	0.108			
session4	-0.381	-0.009	0.379		
activitystanding	-0.415	-0.006	0.410	0.426	
activitywalking	-0.377	0.088	0.129	-0.004	0.000

activitywalking	-0.405	0.033	0.302	-0.004	0.281	0.398	
sexM	-0.485	-0.018	0.006	0.002	0.004	-0.006	0.001
HF_typeHF_2	-0.596	0.284	0.384	0.323	0.352	0.320	0.343
HF_typeHF_3	-0.589	0.282	0.379	0.319	0.347	0.316	0.339
HF_typeHF_5	-0.587	0.280	0.378	0.318	0.346	0.315	0.337
HF_typeHF_7	-0.598	0.286	0.385	0.324	0.353	0.321	0.344
HF_typeHF_8	-0.594	0.283	0.382	0.322	0.350	0.318	0.342
day2:HF_typeHF_2	0.238	-0.710	-0.077	0.006	0.004	-0.062	-0.023
day2:HF_typeHF_3	0.236	-0.706	-0.076	0.006	0.004	-0.062	-0.023
day2:HF_typeHF_5	0.236	-0.702	-0.076	0.006	0.004	-0.061	-0.023
day2:HF_typeHF_7	0.240	-0.713	-0.077	0.007	0.004	-0.062	-0.023
day2:HF_typeHF_8	0.238	-0.708	-0.077	0.006	0.004	-0.062	-0.023
session2:HF_typeHF_2	0.323	-0.077	-0.713	-0.271	-0.293	-0.092	-0.215
session3:HF_typeHF_2	0.271	0.007	-0.270	-0.710	-0.303	0.003	0.003
session4:HF_typeHF_2	0.294	0.004	-0.291	-0.302	-0.709	0.000	-0.199
session2:HF_typeHF_3	0.321	-0.077	-0.708	-0.269	-0.291	-0.091	-0.214
session3:HF_typeHF_3	0.269	0.007	-0.268	-0.707	-0.301	0.003	0.003
session4:HF_typeHF_3	0.292	0.004	-0.288	-0.299	-0.703	0.000	-0.197
session2:HF_typeHF_5	0.319	-0.076	-0.703	-0.267	-0.289	-0.091	-0.213
session3:HF_typeHF_5	0.268	0.007	-0.267	-0.703	-0.299	0.003	0.003
session4:HF_typeHF_5	0.291	0.004	-0.288	-0.299	-0.701	0.000	-0.197
session2:HF_typeHF_7	0.324	-0.077	-0.714	-0.271	-0.293	-0.092	-0.216
session3:HF_typeHF_7	0.272	0.007	-0.271	-0.714	-0.304	0.003	0.003
session4:HF_typeHF_7	0.295	0.004	-0.292	-0.303	-0.711	0.000	-0.200
session2:HF_typeHF_8	0.323	-0.077	-0.713	-0.270	-0.292	-0.092	-0.215
session3:HF_typeHF_8	0.270	0.007	-0.268	-0.707	-0.301	0.003	0.003
session4:HF_typeHF_8	0.294	0.004	-0.291	-0.302	-0.709	0.000	-0.199
activitystanding:HF_typeHF_2	0.268	-0.062	-0.092	0.003	0.000	-0.711	-0.283
activitywalking:HF_typeHF_2	0.287	-0.023	-0.214	0.003	-0.199	-0.282	-0.709
activitystanding:HF_typeHF_3	0.266	-0.062	-0.091	0.003	0.000	-0.706	-0.281
activitywalking:HF_typeHF_3	0.285	-0.023	-0.213	0.003	-0.198	-0.281	-0.705
activitystanding:HF_typeHF_5	0.265	-0.062	-0.091	0.003	0.000	-0.703	-0.280
activitywalking:HF_typeHF_5	0.284	-0.023	-0.212	0.003	-0.197	-0.279	-0.702
activitystanding:HF_typeHF_7	0.269	-0.063	-0.092	0.003	0.000	-0.714	-0.284
activitywalking:HF_typeHF_7	0.289	-0.023	-0.215	0.003	-0.200	-0.284	-0.713
activitystanding:HF_typeHF_8	0.267	-0.062	-0.091	0.003	0.000	-0.709	-0.282
activitywalking:HF_typeHF_8	0.286	-0.023	-0.214	0.003	-0.199	-0.281	-0.707
sexM:HF_typeHF_2	0.209	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_3	0.208	0.020	-0.006	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_5	0.207	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_7	0.210	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM:HF_typeHF_8	0.209	0.020	-0.007	-0.002	-0.005	0.007	-0.001
sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8		
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							
HF_typeHF_2	0.156						
HF_typeHF_3	0.154	0.499					
HF_typeHF_5	0.154	0.497	0.491				
HF_typeHF_7	0.157	0.507	0.501	0.499			

HF_typeHF_8	0.155	0.503	0.497	0.495	0.505	
day2:HF_typeHF_2	0.013	-0.403	-0.200	-0.199	-0.203	-0.202
day2:HF_typeHF_3	0.013	-0.201	-0.411	-0.198	-0.201	-0.200
day2:HF_typeHF_5	0.013	-0.200	-0.198	-0.411	-0.200	-0.199
day2:HF_typeHF_7	0.013	-0.203	-0.201	-0.200	-0.401	-0.202
day2:HF_typeHF_8	0.013	-0.202	-0.200	-0.199	-0.202	-0.404
session2:HF_typeHF_2	-0.004	-0.541	-0.270	-0.269	-0.275	-0.273
session3:HF_typeHF_2	-0.001	-0.458	-0.227	-0.226	-0.230	-0.229
session4:HF_typeHF_2	-0.003	-0.493	-0.246	-0.245	-0.250	-0.248
session2:HF_typeHF_3	-0.004	-0.272	-0.539	-0.268	-0.273	-0.271
session3:HF_typeHF_3	-0.001	-0.228	-0.455	-0.225	-0.229	-0.227
session4:HF_typeHF_3	-0.003	-0.247	-0.491	-0.243	-0.248	-0.246
session2:HF_typeHF_5	-0.004	-0.270	-0.267	-0.540	-0.271	-0.269
session3:HF_typeHF_5	-0.001	-0.227	-0.224	-0.456	-0.228	-0.226
session4:HF_typeHF_5	-0.003	-0.246	-0.243	-0.491	-0.247	-0.246
session2:HF_typeHF_7	-0.004	-0.274	-0.271	-0.270	-0.537	-0.273
session3:HF_typeHF_7	-0.001	-0.231	-0.228	-0.227	-0.457	-0.230
session4:HF_typeHF_7	-0.003	-0.250	-0.247	-0.246	-0.491	-0.249
session2:HF_typeHF_8	-0.004	-0.273	-0.270	-0.269	-0.275	-0.541
session3:HF_typeHF_8	-0.001	-0.228	-0.226	-0.225	-0.229	-0.448
session4:HF_typeHF_8	-0.003	-0.249	-0.246	-0.245	-0.250	-0.493
activitystanding:HF_typeHF_2	0.005	-0.450	-0.225	-0.224	-0.228	-0.227
activitywalking:HF_typeHF_2	-0.001	-0.480	-0.240	-0.239	-0.244	-0.242
activitystanding:HF_typeHF_3	0.004	-0.226	-0.449	-0.222	-0.227	-0.225
activitywalking:HF_typeHF_3	-0.001	-0.242	-0.477	-0.238	-0.243	-0.241
activitystanding:HF_typeHF_5	0.004	-0.224	-0.222	-0.449	-0.225	-0.224
activitywalking:HF_typeHF_5	-0.001	-0.240	-0.237	-0.478	-0.241	-0.240
activitystanding:HF_typeHF_7	0.005	-0.228	-0.226	-0.225	-0.448	-0.227
activitywalking:HF_typeHF_7	-0.001	-0.244	-0.241	-0.241	-0.479	-0.244
activitystanding:HF_typeHF_8	0.004	-0.226	-0.224	-0.223	-0.227	-0.452
activitywalking:HF_typeHF_8	-0.001	-0.242	-0.239	-0.239	-0.243	-0.483
sexM:HF_typeHF_2	-0.444	-0.350	-0.174	-0.174	-0.178	-0.176
sexM:HF_typeHF_3	-0.441	-0.175	-0.358	-0.173	-0.177	-0.175
sexM:HF_typeHF_5	-0.438	-0.175	-0.172	-0.349	-0.176	-0.174
sexM:HF_typeHF_7	-0.446	-0.178	-0.176	-0.175	-0.358	-0.177
sexM:HF_typeHF_8	-0.443	-0.176	-0.174	-0.174	-0.178	-0.353
	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7		
day2						
session2						
session3						
session4						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5						
HF_typeHF_7						
HF_typeHF_8						
day2:HF_typeHF_2						
day2:HF_typeHF_3	0.502					
day2:HF_typeHF_5	0.500	0.496				
day2:HF_typeHF_7	0.507	0.503	0.501			
day2:HF_typeHF_8	0.504	0.501	0.498	0.505		

session2:HF_typeHF_2	0.111	0.054	0.054	0.055
session3:HF_typeHF_2	-0.005	-0.005	-0.004	-0.005
session4:HF_typeHF_2	-0.003	-0.003	-0.003	-0.003
session2:HF_typeHF_3	0.054	0.111	0.054	0.055
session3:HF_typeHF_3	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_3	-0.003	0.001	-0.003	-0.003
session2:HF_typeHF_5	0.054	0.053	0.112	0.054
session3:HF_typeHF_5	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_5	-0.003	-0.003	-0.003	-0.003
session2:HF_typeHF_7	0.055	0.054	0.054	0.112
session3:HF_typeHF_7	-0.005	-0.005	-0.004	-0.005
session4:HF_typeHF_7	-0.003	-0.003	-0.003	-0.001
session2:HF_typeHF_8	0.055	0.054	0.054	0.055
session3:HF_typeHF_8	-0.004	-0.005	-0.004	-0.005
session4:HF_typeHF_8	-0.003	-0.003	-0.003	-0.003
activitystanding:HF_typeHF_2	0.084	0.044	0.044	0.044
activitywalking:HF_typeHF_2	0.032	0.016	0.016	0.017
activitystanding:HF_typeHF_3	0.044	0.085	0.043	0.044
activitywalking:HF_typeHF_3	0.016	0.032	0.016	0.017
activitystanding:HF_typeHF_5	0.044	0.043	0.085	0.044
activitywalking:HF_typeHF_5	0.016	0.016	0.033	0.016
activitystanding:HF_typeHF_7	0.044	0.044	0.044	0.081
activitywalking:HF_typeHF_7	0.017	0.016	0.016	0.031
activitystanding:HF_typeHF_8	0.044	0.044	0.043	0.044
activitywalking:HF_typeHF_8	0.016	0.016	0.016	0.017
sexM:HF_typeHF_2	-0.026	-0.016	-0.015	-0.014
sexM:HF_typeHF_3	-0.015	-0.012	-0.015	-0.014
sexM:HF_typeHF_5	-0.015	-0.016	-0.027	-0.014
sexM:HF_typeHF_7	-0.014	-0.014	-0.014	-0.016
sexM:HF_typeHF_8	-0.016	-0.016	-0.015	-0.014
d2:HF_HF_8 s2:HF_HF_2 s3:HF_HF_2 s4:HF_HF_2				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.055			
session3:HF_typeHF_2	-0.005	0.383		
session4:HF_typeHF_2	-0.003	0.411	0.426	
session2:HF_typeHF_3	0.054	0.505	0.191	0.206
session3:HF_typeHF_3	-0.004	0.191	0.502	0.213
session4:HF_typeHF_3	-0.003	0.205	0.213	0.498

session2:HF_typeHF_5	0.054	0.501	0.190	0.204
session3:HF_typeHF_5	-0.004	0.190	0.499	0.212
session4:HF_typeHF_5	-0.003	0.205	0.212	0.497
session2:HF_typeHF_7	0.055	0.509	0.193	0.208
session3:HF_typeHF_7	-0.005	0.193	0.507	0.216
session4:HF_typeHF_7	-0.003	0.208	0.215	0.504
session2:HF_typeHF_8	0.112	0.508	0.192	0.207
session3:HF_typeHF_8	-0.009	0.191	0.502	0.213
session4:HF_typeHF_8	-0.002	0.207	0.214	0.502
activitystanding:HF_typeHF_2	0.044	0.129	-0.002	0.000
activitywalking:HF_typeHF_2	0.016	0.302	-0.002	0.279
activitystanding:HF_typeHF_3	0.044	0.065	-0.002	0.000
activitywalking:HF_typeHF_3	0.016	0.152	-0.002	0.140
activitystanding:HF_typeHF_5	0.043	0.065	-0.002	0.000
activitywalking:HF_typeHF_5	0.016	0.151	-0.002	0.140
activitystanding:HF_typeHF_7	0.044	0.066	-0.002	0.000
activitywalking:HF_typeHF_7	0.016	0.154	-0.002	0.142
activitystanding:HF_typeHF_8	0.083	0.065	-0.002	0.000
activitywalking:HF_typeHF_8	0.035	0.152	-0.002	0.141
sexM:HF_typeHF_2	-0.016	0.005	0.001	0.004
sexM:HF_typeHF_3	-0.015	0.005	0.001	0.003
sexM:HF_typeHF_5	-0.015	0.005	0.001	0.004
sexM:HF_typeHF_7	-0.014	0.005	0.002	0.003
sexM:HF_typeHF_8	-0.025	0.005	0.001	0.004
s2:HF_HF_3 s3:HF_HF_3 s4:HF_HF_3 s2:HF_HF_5				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.382			
session4:HF_typeHF_3	0.409	0.425		
session2:HF_typeHF_5	0.498	0.189	0.203	
session3:HF_typeHF_5	0.189	0.496	0.210	0.382
session4:HF_typeHF_5	0.204	0.211	0.493	0.410
session2:HF_typeHF_7	0.506	0.192	0.206	0.502
session3:HF_typeHF_7	0.192	0.504	0.214	0.191
session4:HF_typeHF_7	0.207	0.214	0.500	0.205

session2:HF_typeHF_8	0.505	0.191	0.205	0.501
session3:HF_typeHF_8	0.190	0.499	0.212	0.189
session4:HF_typeHF_8	0.206	0.213	0.498	0.204
activitystanding:HF_typeHF_2	0.065	-0.002	0.000	0.064
activitywalking:HF_typeHF_2	0.152	-0.002	0.140	0.151
activitystanding:HF_typeHF_3	0.131	-0.002	0.000	0.064
activitywalking:HF_typeHF_3	0.302	-0.002	0.278	0.150
activitystanding:HF_typeHF_5	0.064	-0.002	0.000	0.132
activitywalking:HF_typeHF_5	0.150	-0.002	0.138	0.303
activitystanding:HF_typeHF_7	0.065	-0.002	0.000	0.065
activitywalking:HF_typeHF_7	0.153	-0.002	0.141	0.152
activitystanding:HF_typeHF_8	0.065	-0.002	0.000	0.064
activitywalking:HF_typeHF_8	0.151	-0.002	0.140	0.150
sexM:HF_typeHF_2	0.005	0.001	0.003	0.005
sexM:HF_typeHF_3	0.005	0.001	0.007	0.005
sexM:HF_typeHF_5	0.005	0.001	0.003	0.005
sexM:HF_typeHF_7	0.005	0.002	0.003	0.005
sexM:HF_typeHF_8	0.005	0.001	0.003	0.005
s3:HF_HF_5 s4:HF_HF_5 s2:HF_HF_7 s3:HF_HF_7				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.426			
session2:HF_typeHF_7	0.190	0.205		
session3:HF_typeHF_7	0.502	0.213	0.381	
session4:HF_typeHF_7	0.213	0.499	0.409	0.426
session2:HF_typeHF_8	0.190	0.205	0.509	0.193
session3:HF_typeHF_8	0.497	0.211	0.192	0.505
session4:HF_typeHF_8	0.212	0.497	0.208	0.216
activitystanding:HF_typeHF_2	-0.002	0.000	0.066	-0.002
activitywalking:HF_typeHF_2	-0.002	0.140	0.153	-0.002
activitystanding:HF_typeHF_3	-0.002	0.000	0.065	-0.002

activitywalking:HF_typeHF_3	-0.002	0.139	0.152	-0.002
activitystanding:HF_typeHF_5	-0.002	0.000	0.065	-0.002
activitywalking:HF_typeHF_5	-0.002	0.279	0.151	-0.002
activitystanding:HF_typeHF_7	-0.002	0.000	0.128	-0.002
activitywalking:HF_typeHF_7	-0.002	0.140	0.301	-0.002
activitystanding:HF_typeHF_8	-0.002	0.000	0.065	-0.002
activitywalking:HF_typeHF_8	-0.002	0.139	0.153	-0.002
sexM:HF_typeHF_2	0.001	0.003	0.005	0.001
sexM:HF_typeHF_3	0.001	0.003	0.005	0.001
sexM:HF_typeHF_5	0.001	0.003	0.005	0.001
sexM:HF_typeHF_7	0.002	0.003	0.000	0.002
sexM:HF_typeHF_8	0.001	0.003	0.005	0.001
s4:HF_HF_7 s2:HF_HF_8 s3:HF_HF_8 s4:HF_HF_8				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.208			
session3:HF_typeHF_8	0.214	0.378		
session4:HF_typeHF_8	0.504	0.411	0.422	
activitystanding:HF_typeHF_2	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_2	0.142	0.153	-0.002	0.141
activitystanding:HF_typeHF_3	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_3	0.141	0.152	-0.002	0.140
activitystanding:HF_typeHF_5	0.000	0.065	-0.002	0.000
activitywalking:HF_typeHF_5	0.140	0.151	-0.002	0.140
activitystanding:HF_typeHF_7	-0.002	0.066	-0.002	0.000
activitywalking:HF_typeHF_7	0.278	0.154	-0.002	0.142
activitystanding:HF_typeHF_8	0.000	0.130	-0.004	0.001

activitywalking:HF_typeHF_8	0.141	0.303	-0.009	0.280
sexM:HF_typeHF_2	0.004	0.005	0.001	0.004
sexM:HF_typeHF_3	0.003	0.005	0.001	0.003
sexM:HF_typeHF_5	0.004	0.005	0.002	0.003
sexM:HF_typeHF_7	0.002	0.005	0.002	0.003
sexM:HF_typeHF_8	0.004	0.006	-0.003	0.004
	actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3	
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2	0.396			
activitystanding:HF_typeHF_3	0.502	0.199		
activitywalking:HF_typeHF_3	0.200	0.500		0.397
activitystanding:HF_typeHF_5	0.500	0.198		0.496
activitywalking:HF_typeHF_5	0.199	0.498		0.197
activitystanding:HF_typeHF_7	0.508	0.201		0.504
activitywalking:HF_typeHF_7	0.202	0.506		0.200
activitystanding:HF_typeHF_8	0.504	0.200		0.500
activitywalking:HF_typeHF_8	0.200	0.502		0.199
sexM:HF_typeHF_2	-0.005	0.001		-0.005
sexM:HF_typeHF_3	-0.005	0.001		-0.005
sexM:HF_typeHF_5	-0.005	0.001		-0.005
sexM:HF_typeHF_7	-0.005	0.001		-0.005
sexM:HF_typeHF_8	-0.005	0.001		-0.005

```

actvttyw:HF_HF_3 actvtys:HF_HF_5 actvttyw:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5 0.197
activitywalking:HF_typeHF_5 0.495 0.396
activitystanding:HF_typeHF_7 0.200 0.502 0.199
activitywalking:HF_typeHF_7 0.503 0.199 0.500
activitystanding:HF_typeHF_8 0.199 0.498 0.198
activitywalking:HF_typeHF_8 0.499 0.198 0.496
sexM:HF_typeHF_2 0.001 -0.005 0.001
sexM:HF_typeHF_3 0.001 -0.005 0.001
sexM:HF_typeHF_5 0.001 -0.005 0.001
sexM:HF_typeHF_7 0.001 -0.005 0.001
sexM:HF_typeHF_8 0.001 -0.005 0.001
actvtys:HF_HF_7 actvttyw:HF_HF_7 actvtys:HF_HF_8

day2
session2
session3
session4
activitystanding

```

```

activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7    0.396
activitystanding:HF_typeHF_8    0.506            0.201
activitywalking:HF_typeHF_8    0.201            0.504            0.397
sexM:HF_typeHF_2                -0.005            0.001            -0.005
sexM:HF_typeHF_3                -0.005            0.001            -0.005
sexM:HF_typeHF_5                -0.005            0.001            -0.005
sexM:HF_typeHF_7                -0.005            0.001            -0.005
sexM:HF_typeHF_8                -0.005            0.001            -0.001
                                actvttyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7

```

```

HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2      0.001
sexM:HF_typeHF_3      0.001      0.502
sexM:HF_typeHF_5      0.001      0.500      0.496
sexM:HF_typeHF_7      0.001      0.507      0.503      0.501
sexM:HF_typeHF_8      0.005      0.504      0.501      0.498
                                sM:HF_HF_7

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8

```

```

session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.505

```

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-6.4785480	-0.4907863	0.1117616	0.6319402	3.7636443

Number of Observations: 2638

Number of Groups: 24

```
> anova(m0_gls_reml, m1)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
m0_gls_reml	1	49	1355.773	1642.885	-628.8867			
m1	2	50	1207.812	1500.783	-553.9061	1 vs 2	149.9612	<.0001

```
> print(VarCorr(m1))
```

```

person = pdLogChol(1)
      Variance StdDev
(Intercept) 0.006945276 0.08333832
Residual    0.080883753 0.28440069

```

```

> mdl4a_reml <- m1 #align naming conventions
> print(VarCorr(mdl4a_reml))

```

```

person = pdLogChol(1)
      Variance StdDev

```

```

(Intercept) 0.006945276 0.08333832
Residual    0.080883753 0.28440069

> mdl4a_var <- as.numeric(VarCorr(mdl4a_reml))
> print(mdl4a_var )

[1] 0.006945276 0.080883753 0.083338320 0.284400690

> #these are the variances
> print(mdl4a_var[1])

[1] 0.006945276

> print(mdl4a_var[2])

[1] 0.08088375

> mdl4a_var_person <- mdl4a_var[1]
> mdl4a_var_residual <- mdl4a_var[2]
>
> mdl4a_icc <- mdl4a_var_person/(mdl4a_var_person+mdl4a_var_residual)
> mdl4a_icc_res <- mdl4a_var_residual/(mdl4a_var_person+mdl4a_var_residual)
> print(mdl4a_icc)

[1] 0.07907723

> print(mdl4a_icc_res)

[1] 0.9209228

> mdl3_reml <- m0_gls_reml #align naming conventions
>
> anova_mdl3_mdl4 <- bind_rows(
+   get_baseline_metrics(mdl3_reml, "Fixed Effect Model"),
+   get_anova_outputs(mdl3_reml, mdl4a_reml, "Person Random Intercept Model")
+ )
>
> anova_mdl3_mdl4_cleaned <- anova_mdl3_mdl4 %>%
+   mutate(
+     LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
+     p_value = ifelse(is.na(p_value), "--",
+       # eps defines the threshold for "<"
+       # digits defines the rounding for values above that threshold
+       format.pval(p_value, eps = 0.0001, digits = 4))
+   )
>
>
> col_labels <- c("Model Version",
+   "\\shortstack{Degrees of\\\\Freedom}",
+   "AIC",
+   "BIC",
+   "Log-Likelihood",
+   "\\shortstack{Likelihood Ratio\\\\Test Stat}",
+   "P-Value")

```

To determine if there is inter-individual variability, we added a random intercept for each person. This assumes that all individuals have unique baselines for heat flow, regardless of if they are performing the same activity and/or are at the same temperature. Note the models in this section were run using REML to ensure estimates of variance components are not biased, unless explicitly stated otherwise. The Intra-

class Correlation Coefficient (ICC) (also known to as intra-individual coefficient here) for this model was 0.0791, implying that ~8% of the variance is due to differences between people (and 92% is unexplained). This suggests there is some inter-individual variability. Table 1 shows that this also reduces AIC and BIC significantly.

```
> kable(anova_md13_md14_cleaned,
+       format="latex",
+       col.names = col_labels,
+       align = "lrrrrrr",
+       booktabs = TRUE, #, #right aligns numbers
+       escape = FALSE,
+       caption = "\\footnotesize{Impact of adding a random intercept for person to the fixed-effect on",
+       kable_styling(latex_options = c("scale_down", "hold_position"),
+       full_width = FALSE,
+       table.envir = "table") %>%
+       add_header_above(c("Likelihood Ratio Test for Random Intercept for Person" = 7),
+       bold = TRUE, align = "l") %>%
+       row_spec(0, bold = TRUE)
```

Table 1: Impact of adding a random intercept for person to the fixed-effect only model. Note this uses REML to compare models, as fixed effects are not changing. As such, the final p-values should be divided by 2. Given the p-value is already extremely low, this has no impact on the takeaway that we should include the Random Intercept for Person. Note models were fit using Restricted Maximum Likelihood (REML), as fixed effects are the same between models.

Likelihood Ratio Test for Random Intercept for Person						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Fixed Effect Model	49	1356	1643	-629	–	–
Person Random Intercept Model	50	1208	1501	-554	149.96	< 1e-04

Beyond inter-individual variability, we hypothesized that heat flow for a given activity may vary uniquely within each individual. A nested random-effects structure ($\sim 1 \mid \text{person/activity}$) was tested; we found that the ICC for person remained the same as in model 4 (0.0791) and the ICC of activity (within person) is 0, implying that once fixed effects for activity are accounted for there is no additional person-specific variation across different activities. As such, we did not include activity as a nested random intercept.

```
> m2 <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+          random = ~ 1 | person/activity,
+          data = data_clean,
+          method = "REML")
>
> #summary(m2)
>
> print(VarCorr(m2))
```

```

      Variance      StdDev
person = pdLogChol(1)
(Intercept) 6.945276e-03 8.333832e-02
activity = pdLogChol(1)
(Intercept) 2.265352e-10 1.505109e-05
Residual    8.088375e-02 2.844007e-01
```

```
> mdl4b_reml <- m2 #rename for ease of paste code
>
> print(VarCorr(mdl4b_reml))
```

```

      Variance      StdDev
person = pdLogChol(1)
(Intercept) 6.945276e-03 8.333832e-02
activity = pdLogChol(1)
(Intercept) 2.265352e-10 1.505109e-05
Residual    8.088375e-02 2.844007e-01

> mdl4b_var <- as.numeric(VarCorr(mdl4b_reml))
> print(mdl4b_var )

[1]      NA 6.945276e-03      NA 2.265352e-10 8.088375e-02
[6]      NA 8.333832e-02      NA 1.505109e-05 2.844007e-01

> #these are the variances
> print(mdl4b_var[1])

[1] NA
> print(mdl4b_var[2])

[1] 0.006945276
> print(mdl4b_var[3])

[1] NA
> print(mdl4b_var[4])

[1] 2.265352e-10
> print(mdl4b_var[5])

[1] 0.08088375
> print(mdl4b_var[6])

[1] NA
> mdl4b_var_person <- mdl4b_var[2]
> mdl4b_var_activity <- mdl4b_var[4]
> mdl4b_var_residual <- mdl4b_var[5]
> mdl4b_var_total <- mdl4b_var_person + mdl4b_var_activity + mdl4b_var_residual
>
> mdl4b_icc_person <- round(mdl4b_var_person / mdl4b_var_total, 4)
> mdl4b_icc_activity <- round(mdl4b_var_activity / mdl4b_var_total, 4)
> mdl4b_icc_res <- round(mdl4b_var_residual / mdl4b_var_total, 4)
>
> print("icc person")

[1] "icc person"
> print(mdl4b_icc_person)

[1] 0.0791
> print("icc activity")

[1] "icc activity"
> print(mdl4b_icc_activity)

[1] 0

```

```
> print("icc residual")
```

```
[1] "icc residual"
```

```
> print mdl4b_icc_res)
```

```
[1] 0.9209
```

We further hypothesized that baseline heat flow differs based on sensor location for each individual, so we tested the nested random-effects structure ($\sim 1 \mid \text{person/sensor}$). This resulted in a model with 60% of the variance left unexplained (compared to the previous model's 92%); the ICC for person decreased from the person-only random intercept model (from 0.0791 to 0.0184), and the ICC of sensor (within person) was 0.3772. This implies that the major source of clustering is at the sensor level within a person.

Removing the person random intercept (resulting in a sensor-only random intercept model) results in the significantly lower ICC for sensor (0.0159), with 98% of the variance left unexplained. This implies that the sensor by itself is not the primary driver of variance; instead, there is a shared baseline amongst sensors belonging to the same individual. Thus, we kept the nested sensor-level random intercept model.

```
> m2_HFtype <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+                 random = ~ 1 | person/HF_type,
+                 data = data_clean,
+                 method = "REML")
>
> #summary(m2_HFtype)
```

```
> print("hi")
```

```
[1] "hi"
```

```
> print(VarCorr(m2_HFtype))
```

	Variance	StdDev
person =	pdLogChol(1)	
(Intercept)	0.001667335	0.04083301
HF_type =	pdLogChol(1)	
(Intercept)	0.034177320	0.18487109
Residual	0.054767267	0.23402407

```
> anova(m1, m2_HFtype)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
m1		1 50	1207.812	1500.783	-553.9061			
m2_HFtype		2 51	484.768	783.598	-191.3840	1 vs 2	725.0442	<.0001

```
> mdl4c_reml <- m2_HFtype #rename for easier printing later
```

```
>
```

```
>
```

```
> print(VarCorr(mdl4c_reml))
```

	Variance	StdDev
person =	pdLogChol(1)	
(Intercept)	0.001667335	0.04083301
HF_type =	pdLogChol(1)	
(Intercept)	0.034177320	0.18487109
Residual	0.054767267	0.23402407

```
> mdl4c_var <- as.numeric(VarCorr(mdl4c_reml))
```

```
> print(mdl4c_var )
```

```

[1] NA 0.001667335 NA 0.034177320 0.054767267 NA
[7] 0.040833010 NA 0.184871090 0.234024070

> #these are the variances
> print(mdl4c_var[1])

[1] NA

> print(mdl4c_var[2])

[1] 0.001667335

> print(mdl4c_var[3])

[1] NA

> print(mdl4c_var[4])

[1] 0.03417732

> print(mdl4c_var[5])

[1] 0.05476727

> print(mdl4c_var[6])

[1] NA

> mdl4c_var_person <- mdl4c_var[2]
> mdl4c_var_sensor <- mdl4c_var[4]
> mdl4c_var_residual <- mdl4c_var[5]
> mdl4c_var_total <- mdl4c_var_person + mdl4c_var_sensor + mdl4c_var_residual
>
> mdl4c_icc_person <- round(mdl4c_var_person / mdl4c_var_total, 4)
> mdl4c_icc_sensor <- round(mdl4c_var_sensor / mdl4c_var_total, 4)
> mdl4c_icc_res <- round(mdl4c_var_residual / mdl4c_var_total, 4)
>
> print("icc person")

[1] "icc person"

> print(mdl4c_icc_person)

[1] 0.0184

> print("icc sensor")

[1] "icc sensor"

> print(mdl4c_icc_sensor)

[1] 0.3772

> print("icc residual")

[1] "icc residual"

> print(mdl4c_icc_res)

[1] 0.6044

> mdl4d_reml <- lme(fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+                  random = ~ 1 | HF_type, #unique intercept for each person

```

```

+         data = data_clean,
+         method = "REML")
>
> print(VarCorr(mdl4d_reml))

HF_type = pdLogChol(1)
      Variance   StdDev
(Intercept) 0.001414016 0.03760341
Residual    0.087425966 0.29567882
> mdl4d_var <- as.numeric(VarCorr(mdl4d_reml))
> print(mdl4d_var )

[1] 0.001414016 0.087425966 0.037603410 0.295678820
> #these are the variances
> print(mdl4d_var[1])

[1] 0.001414016
> print(mdl4d_var[2])

[1] 0.08742597
> print(mdl4d_var[3])

[1] 0.03760341
> print(mdl4d_var[4])

[1] 0.2956788
> mdl4d_var_sensor <- mdl4d_var[1]
> mdl4d_var_residual <- mdl4d_var[2]
> mdl4d_var_total <- mdl4d_var_sensor + mdl4d_var_residual
>
> mdl4d_icc_sensor <- round(mdl4d_var_sensor / mdl4d_var_total, 4)
> mdl4d_icc_res <- round(mdl4d_var_residual / mdl4d_var_total, 4)
>
> print("icc sensor")

[1] "icc sensor"
> print(mdl4d_icc_sensor)

[1] 0.0159
> print("icc residaul")

[1] "icc residaul"
> print(mdl4d_icc_res)

[1] 0.9841

```

5.3 Random Slope

The current model has random intercepts for person and sensor (nested), implying that measurements from the same sensor are correlated, and sensors belonging to the same person are correlated. However, it assumes that the impact of temperature (the day variable) is identical for all individuals. We tested day as a random

slope (given person) to determine if the effect of day varies by person, meaning some people adapt to the temperature environment differently than others.

```
> m3 <- lme(
+   fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   data = data_clean,
+   method = "REML"
+ )
>
> anova(m2_HFtype, m3)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
m2_HFtype	1	51	484.768	783.5980	-191.384			
m3	2	53	311.948	622.4968	-102.974	1 vs 2	176.82	<.0001

```
> print(VarCorr(m3))
```

	Variance	StdDev	Corr
person = pdLogChol(1 + day)			
(Intercept)	0.004649289	0.0681857	(Intr)
day2	0.020644014	0.1436803	-0.884
HF_type = pdLogChol(1)			
(Intercept)	0.035971708	0.1896621	
Residual	0.049731894	0.2230065	

```
> mdl6_reml <- m3 #rename
>
> print(anova(mdl4c_reml, mdl6_reml))
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl4c_reml	1	51	484.768	783.5980	-191.384			
mdl6_reml	2	53	311.948	622.4968	-102.974	1 vs 2	176.82	<.0001

```
> print(VarCorr(mdl6_reml))
```

	Variance	StdDev	Corr
person = pdLogChol(1 + day)			
(Intercept)	0.004649289	0.0681857	(Intr)
day2	0.020644014	0.1436803	-0.884
HF_type = pdLogChol(1)			
(Intercept)	0.035971708	0.1896621	
Residual	0.049731894	0.2230065	

```
> mdl6_var <- as.numeric(VarCorr(mdl6_reml))
> print(mdl6_var)
```

[1]	NA	0.004649289	0.020644014	NA	0.035971708
[6]	0.049731894	NA	0.068185700	0.143680300	NA
[11]	0.189662100	0.223006500	NA	NA	-0.884000000
[16]	NA	NA	NA		

```
> #these are the variances
> print(mdl6_var[1])
```

```

[1] NA
> print(mdl6_var[2])

[1] 0.004649289
> print(mdl6_var[3])

[1] 0.02064401
> print(mdl6_var[4])

[1] NA
> print(mdl6_var[5])

[1] 0.03597171
> print(mdl6_var[6])

[1] 0.04973189
> print(mdl6_var[15]) #correlation

[1] -0.884
> mdl6_var_person <- mdl6_var[2]
> mdl6_var_day <- mdl6_var[3]
> mdl6_var_sensor <- mdl6_var[5]
> mdl6_var_residual <- mdl6_var[6]
> mdl6_var_total <- mdl6_var_person + mdl6_var_sensor + mdl6_var_day+ mdl6_var_residual
>
> mdl6_icc_person <- round(mdl6_var_person / mdl6_var_total, 4)
> mdl6_icc_sensor <- round(mdl6_var_sensor / mdl6_var_total, 4)
> mdl6_icc_day <- round(mdl6_var_day / mdl6_var_total, 4)
> mdl6_icc_res <- round(mdl6_var_residual / mdl6_var_total, 4)
>
> mdl6_corr_day_person <- mdl6_var[15]
>
> print("icc person mdl6")

[1] "icc person mdl6"
> print(mdl6_icc_person)

[1] 0.0419
> print("icc day mdl6")

[1] "icc day mdl6"
> print(mdl6_icc_day)

[1] 0.186
> print("icc sensor mdl6")

[1] "icc sensor mdl6"
> print(mdl6_icc_sensor)

[1] 0.3241

```

```
> print("icc residual mdl6")
```

```
[1] "icc residual mdl6"
```

```
> print(mdl6_icc_res)
```

```
[1] 0.448
```

```
> print("icc person mdl4c")
```

```
[1] "icc person mdl4c"
```

```
> print(mdl4c_icc_person)
```

```
[1] 0.0184
```

```
> print("icc sensor mdl4c")
```

```
[1] "icc sensor mdl4c"
```

```
> print(mdl4c_icc_sensor)
```

```
[1] 0.3772
```

```
> print("icc residual mdl4c")
```

```
[1] "icc residual mdl4c"
```

```
> print(mdl4c_icc_res)
```

```
[1] 0.6044
```

```
> print("corr day person")
```

```
[1] "corr day person"
```

```
> print(mdl6_corr_day_person)
```

```
[1] -0.884
```

Adding day as a random slope results in 45% unexplained variance, compared to 60% from the previous model (without a random slope). The ICC for day is 0.186, meaning ~19% of the variation in heat flow measurements is explained by the fact that people have different heat flow reactions depending on the external temperature. Additionally, the correlation between the person-level intercept and the day-level random slope is -0.884. This implies that the higher a person's heat flow was on day 1 (hot day), the steeper their drop was on day 2 (cold day), suggesting convergence in heat flow.

The ICC for person also increased to 0.0419 (compared to 0.0184 previously), suggesting that the noise from people reacting differently to temperature masked their baseline differences. The ICC for sensor decreased to 0.3241 (from 0.3772), suggesting that temperature explained some of the heat flow variance that was previously attributed to sensor.

```
> anova_md16 <- anova(mdl4c_reml, mdl6_reml)
> anova_md16_clean <- data.frame(
+   Model = c("Random Intercept: Person and Nested Sensor",
+             "Adds Random Slope for Day (Given Person)"),
+   Df = as.numeric(anova_md16$df),
+   AIC = round(as.numeric(anova_md16$AIC)),
+   BIC = round(as.numeric(anova_md16$BIC)),
+   LogLik = round(as.numeric(anova_md16$logLik)),
+   LRT_Stat = c(NA, round(anova_md16$L.Ratio[2], 2)),
+   # FIX: Pass a vector of TWO values instead of just one
```

```

+   p_value = c(NA, anova_md16$p-value[2])
+ )
>
> print(anova_md16)

      Model df      AIC      BIC   logLik   Test L.Ratio p-value
mdl4c_reml   1 51 484.768 783.5980 -191.384
mdl6_reml    2 53 311.948 622.4968 -102.974 1 vs 2 176.82 <.0001

> anova_md16_clean <- anova_md16_clean %>%
+   mutate(
+     LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
+     p_value = ifelse(is.na(p_value), "--",
+       # eps defines the threshold for "<"
+       # digits defines the rounding for values above that threshold
+       format.pval(p_value, eps = 0.0001, digits = 4))
+   )
>
> col_labels <- c("Model Version",
+   "\\shortstack{Degrees of\\\\Freedom}",
+   "AIC",
+   "BIC",
+   "Log-Likelihood",
+   "\\shortstack{Likelihood Ratio\\\\Test Stat}",
+   "P-Value")

> kable(anova_md16_clean,
+   format="latex",
+   col.names = col_labels,
+   align = "lrrrrrr",
+   booktabs = TRUE, #, #right aligns numbers
+   escape = FALSE,
+   caption = "\\footnotesize{Likelihood test between the model with the random intercepts for person and sensor (nested) only, and the model that adds a random slope for day to that model. The model that adds the random slope for day has significantly lower AIC and BIC, and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.}",
+   kable_styling(latex_options = c("scale_down", "hold_position"),
+     full_width = FALSE,
+     table.envir = "table") %>%
+   add_header_above(c("Likelihood Ratio Test for Random Slope for day" = 7),
+     bold = TRUE, align = "l") %>%
+   row_spec(0, bold = TRUE)

```

Table 2: Likelihood test between the model with the random intercepts for person and sensor (nested) only, and the model that adds a random slope for day to that model. The model that adds the random slope for day has significantly lower AIC and BIC, and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Random Slope for day							
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value	
Random Intercept: Person and Nested Sensor	51	485	784	-191	—	—	
Adds Random Slope for Day (Given Person)	53	312	622	-103	176.82	< 1e-04	

5.4 Covariance Structure AR(1)

The selected hierarchical structure effectively partitions the data's variance into three distinct components: global participant-level baselines, individual thermal adaptation patterns (represented by the random slope for day), and nested sensor-level deviations. This framework accounts for the inherent dependencies in the data while acknowledging that heat flow sensitivity is not uniform across the study population.

However, this still imposes a strict assumption of conditional independence for the remaining residual errors. It has a compound symmetry structure, assuming a constant correlation structure where the correlation between session 1 and session 2 is identical to the correlation between session 1 and session 4, ignoring the fact that observations closer in time are typically more highly correlated. As such, we examined implementing an autoregressive (AR1) correlation structure for the residuals. We assumed this was nested within person, sensor, and day, ensuring independence between people, sensor location, and days. We did not assume independence between activity because the activities were done in order, with session 1 consisting of sitting, then standing, then walking, then a break, and then session two beginning with the same cycle (although sessions 2 and 4 did not have walking). This assumes that these measurements were equally spaced in time, which is likely not true - timestamp data could improve this assumption.

The correlation structure has a ρ of 0.5814, meaning that the correlation between session 1 and 2 is ~ 0.58 , and session 1 & 3 is $\sim 0.58^2$. This is fairly high, suggesting the previous data point strongly influences the next one. Since we have the data segmented by session and activity, it suggests that sitting in session 1 impacts standing in session 1, and walking in session 1 impacts sitting in session 2. Additionally, Table 3 shows that the AR1 model has significantly lower AIC and BIC, and is statistically significant in the likelihood ratio test against the previous model with compound symmetry, suggesting an AR1 covariance structure is a significant improvement on the Compound Symmetry structure, and that there is a strong temporal auto-correlation.

```
> m3_ar1 <- lme(
+   fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   data = data_clean,
+   method = "REML"
+ )
>
> anova(m3, m3_ar1)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
m3	1	53	311.9480	622.4968	-102.9740			
m3_ar1	2	54	-402.5175	-86.1092	255.2587	1 vs 2	716.4654	<.0001

```
> summary(m3_ar1)
```

Linear mixed-effects model fit by REML

Data: data_clean

	AIC	BIC	logLik
	-402.5175	-86.10916	255.2587

Random effects:

Formula: ~1 + day | person

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
--	--------	------

(Intercept)	0.06228628	(Intr)
-------------	------------	--------

day2	0.12021252	-0.753
------	------------	--------

Formula: ~1 | HF_type %in% person
 (Intercept) Residual
 StdDev: 0.1683374 0.2434768

Correlation Structure: ARMA(1,0)

Formula: ~time_step | person/HF_type/day
 Parameter estimate(s):

Phi1
 0.581428

Fixed effects: log(HF_value) ~ (day + session + activity + sex) * HF_type

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.492092	0.06620644	2458	67.84978	0.0000
day2	-0.151090	0.04686440	2458	-3.22399	0.0013
session2	-0.060710	0.03335849	2458	-1.81993	0.0689
session3	-0.093293	0.03552884	2458	-2.62585	0.0087
session4	-0.225960	0.04387847	2458	-5.14969	0.0000
activitystanding	-0.090466	0.01625677	2458	-5.56479	0.0000
activitywalking	0.038162	0.02369563	2458	1.61052	0.1074
sexM	0.000141	0.08125815	22	0.00174	0.9986
HF_typeHF_2	-0.720829	0.09102691	110	-7.91886	0.0000
HF_typeHF_3	-0.628960	0.09192644	110	-6.84199	0.0000
HF_typeHF_5	-0.425824	0.09204280	110	-4.62637	0.0000
HF_typeHF_7	-0.170371	0.09087700	110	-1.87474	0.0635
HF_typeHF_8	-0.914155	0.09106227	110	-10.03879	0.0000
day2:HF_typeHF_2	0.479110	0.05641117	2458	8.49317	0.0000
day2:HF_typeHF_3	0.467465	0.05701371	2458	8.19917	0.0000
day2:HF_typeHF_5	0.415906	0.05758430	2458	7.22255	0.0000
day2:HF_typeHF_7	0.313506	0.05587186	2458	5.61116	0.0000
day2:HF_typeHF_8	0.466881	0.05643508	2458	8.27289	0.0000
session2:HF_typeHF_2	0.101432	0.04686735	2458	2.16423	0.0305
session3:HF_typeHF_2	0.114168	0.05010225	2458	2.27869	0.0228
session4:HF_typeHF_2	0.309932	0.06191966	2458	5.00539	0.0000
session2:HF_typeHF_3	0.034543	0.04713517	2458	0.73284	0.4637
session3:HF_typeHF_3	0.075770	0.05037858	2458	1.50402	0.1327
session4:HF_typeHF_3	0.209828	0.06240009	2458	3.36263	0.0008
session2:HF_typeHF_5	0.069928	0.04740210	2458	1.47522	0.1403
session3:HF_typeHF_5	0.087263	0.05064438	2458	1.72306	0.0850
session4:HF_typeHF_5	0.226654	0.06259739	2458	3.62082	0.0003
session2:HF_typeHF_7	0.034441	0.04676978	2458	0.73639	0.4616
session3:HF_typeHF_7	0.037651	0.04986523	2458	0.75505	0.4503
session4:HF_typeHF_7	0.101500	0.06161914	2458	1.64722	0.0996
session2:HF_typeHF_8	0.070919	0.04692627	2458	1.51129	0.1308
session3:HF_typeHF_8	0.153899	0.05020198	2458	3.06561	0.0022
session4:HF_typeHF_8	0.265800	0.06192747	2458	4.29212	0.0000
activitystanding:HF_typeHF_2	-0.088562	0.02277661	2458	-3.88831	0.0001
activitywalking:HF_typeHF_2	-0.377455	0.03333908	2458	-11.32171	0.0000
activitystanding:HF_typeHF_3	0.190024	0.02295586	2458	8.27782	0.0000
activitywalking:HF_typeHF_3	0.220317	0.03354829	2458	6.56717	0.0000
activitystanding:HF_typeHF_5	0.064309	0.02307191	2458	2.78734	0.0054
activitywalking:HF_typeHF_5	-0.049959	0.03372815	2458	-1.48121	0.1387
activitystanding:HF_typeHF_7	-0.032606	0.02270219	2458	-1.43626	0.1511
activitywalking:HF_typeHF_7	-0.117825	0.03319076	2458	-3.54995	0.0004
activitystanding:HF_typeHF_8	0.176924	0.02290290	2458	7.72495	0.0000

activitywalking:HF_typeHF_8	0.145757	0.03345083	2458	4.35735	0.0000		
sexM:HF_typeHF_2	-0.082020	0.11235763	110	-0.72999	0.4669		
sexM:HF_typeHF_3	0.062853	0.11264986	110	0.55795	0.5780		
sexM:HF_typeHF_5	0.017048	0.11291341	110	0.15098	0.8803		
sexM:HF_typeHF_7	0.129723	0.11206504	110	1.15757	0.2495		
sexM:HF_typeHF_8	0.086832	0.11236764	110	0.77275	0.4413		
Correlation:							
	(Intr)	day2	sessn2	sessn3	sessn4	actvtys	actvtyw
day2	-0.321						
session2	-0.277	0.027					
session3	-0.306	-0.004	0.577				
session4	-0.341	-0.010	0.527	0.665			
activitystanding	-0.167	0.016	0.091	0.077	0.130		
activitywalking	-0.230	0.005	0.394	0.156	0.449	0.404	
sexM	-0.603	-0.018	0.003	0.003	0.003	-0.001	0.002
HF_typeHF_2	-0.689	0.179	0.202	0.222	0.248	0.121	0.167
HF_typeHF_3	-0.682	0.177	0.200	0.220	0.246	0.120	0.166
HF_typeHF_5	-0.681	0.177	0.199	0.220	0.245	0.120	0.165
HF_typeHF_7	-0.690	0.179	0.202	0.223	0.249	0.121	0.167
HF_typeHF_8	-0.689	0.179	0.201	0.222	0.248	0.121	0.167
day2:HF_typeHF_2	0.204	-0.603	-0.022	0.004	0.009	-0.013	-0.004
day2:HF_typeHF_3	0.202	-0.597	-0.022	0.003	0.008	-0.013	-0.004
day2:HF_typeHF_5	0.200	-0.591	-0.022	0.003	0.009	-0.013	-0.004
day2:HF_typeHF_7	0.206	-0.609	-0.023	0.004	0.009	-0.014	-0.004
day2:HF_typeHF_8	0.204	-0.603	-0.022	0.004	0.009	-0.013	-0.004
session2:HF_typeHF_2	0.197	-0.019	-0.712	-0.411	-0.375	-0.065	-0.281
session3:HF_typeHF_2	0.217	0.003	-0.409	-0.709	-0.472	-0.055	-0.110
session4:HF_typeHF_2	0.242	0.007	-0.373	-0.471	-0.709	-0.092	-0.318
session2:HF_typeHF_3	0.196	-0.019	-0.708	-0.408	-0.373	-0.064	-0.279
session3:HF_typeHF_3	0.216	0.003	-0.407	-0.705	-0.469	-0.054	-0.110
session4:HF_typeHF_3	0.240	0.007	-0.371	-0.468	-0.703	-0.091	-0.316
session2:HF_typeHF_5	0.195	-0.019	-0.704	-0.406	-0.371	-0.064	-0.278
session3:HF_typeHF_5	0.214	0.003	-0.405	-0.702	-0.467	-0.054	-0.109
session4:HF_typeHF_5	0.239	0.007	-0.369	-0.466	-0.701	-0.091	-0.315
session2:HF_typeHF_7	0.198	-0.019	-0.713	-0.411	-0.376	-0.065	-0.281
session3:HF_typeHF_7	0.218	0.003	-0.411	-0.712	-0.474	-0.055	-0.111
session4:HF_typeHF_7	0.243	0.007	-0.375	-0.474	-0.712	-0.093	-0.320
session2:HF_typeHF_8	0.197	-0.019	-0.711	-0.410	-0.375	-0.064	-0.280
session3:HF_typeHF_8	0.216	0.003	-0.408	-0.708	-0.471	-0.055	-0.110
session4:HF_typeHF_8	0.242	0.007	-0.373	-0.471	-0.709	-0.092	-0.318
activitystanding:HF_typeHF_2	0.119	-0.012	-0.065	-0.055	-0.093	-0.714	-0.288
activitywalking:HF_typeHF_2	0.163	-0.003	-0.280	-0.111	-0.319	-0.287	-0.711
activitystanding:HF_typeHF_3	0.118	-0.011	-0.064	-0.055	-0.092	-0.708	-0.286
activitywalking:HF_typeHF_3	0.162	-0.003	-0.279	-0.110	-0.317	-0.285	-0.706
activitystanding:HF_typeHF_5	0.117	-0.011	-0.064	-0.054	-0.092	-0.705	-0.285
activitywalking:HF_typeHF_5	0.161	-0.003	-0.277	-0.109	-0.315	-0.284	-0.703
activitystanding:HF_typeHF_7	0.119	-0.012	-0.065	-0.055	-0.093	-0.716	-0.289
activitywalking:HF_typeHF_7	0.164	-0.003	-0.282	-0.111	-0.321	-0.288	-0.714
activitystanding:HF_typeHF_8	0.118	-0.011	-0.064	-0.055	-0.092	-0.710	-0.287
activitywalking:HF_typeHF_8	0.163	-0.003	-0.279	-0.110	-0.318	-0.286	-0.708
sexM:HF_typeHF_2	0.417	0.013	-0.002	-0.002	-0.002	0.001	-0.002
sexM:HF_typeHF_3	0.416	0.013	-0.002	-0.002	-0.002	0.001	-0.002
sexM:HF_typeHF_5	0.415	0.013	-0.002	-0.002	-0.002	0.001	-0.002
sexM:HF_typeHF_7	0.418	0.013	-0.002	-0.002	-0.002	0.001	-0.001

sexM:HF_typeHF_8	0.417	0.013	-0.002	-0.002	-0.002	0.001	-0.002
	sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8	
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							
HF_typeHF_2	0.419						
HF_typeHF_3	0.415	0.496					
HF_typeHF_5	0.415	0.496	0.491				
HF_typeHF_7	0.420	0.502	0.497	0.496			
HF_typeHF_8	0.419	0.501	0.496	0.495	0.502		
day2:HF_typeHF_2	0.015	-0.298	-0.148	-0.148	-0.149	-0.149	
day2:HF_typeHF_3	0.015	-0.148	-0.312	-0.146	-0.147	-0.148	
day2:HF_typeHF_5	0.014	-0.146	-0.144	-0.314	-0.146	-0.146	
day2:HF_typeHF_7	0.015	-0.150	-0.148	-0.148	-0.296	-0.150	
day2:HF_typeHF_8	0.015	-0.149	-0.148	-0.148	-0.149	-0.299	
session2:HF_typeHF_2	-0.002	-0.284	-0.142	-0.142	-0.144	-0.143	
session3:HF_typeHF_2	-0.002	-0.314	-0.156	-0.156	-0.158	-0.158	
session4:HF_typeHF_2	-0.002	-0.349	-0.174	-0.174	-0.176	-0.176	
session2:HF_typeHF_3	-0.002	-0.143	-0.283	-0.141	-0.143	-0.143	
session3:HF_typeHF_3	-0.002	-0.157	-0.313	-0.155	-0.157	-0.157	
session4:HF_typeHF_3	-0.002	-0.175	-0.349	-0.173	-0.175	-0.174	
session2:HF_typeHF_5	-0.002	-0.142	-0.141	-0.284	-0.142	-0.142	
session3:HF_typeHF_5	-0.002	-0.156	-0.155	-0.314	-0.156	-0.156	
session4:HF_typeHF_5	-0.002	-0.174	-0.173	-0.349	-0.174	-0.174	
session2:HF_typeHF_7	-0.002	-0.144	-0.143	-0.142	-0.282	-0.144	
session3:HF_typeHF_7	-0.002	-0.158	-0.157	-0.157	-0.313	-0.158	
session4:HF_typeHF_7	-0.002	-0.177	-0.175	-0.175	-0.348	-0.177	
session2:HF_typeHF_8	-0.002	-0.143	-0.142	-0.142	-0.144	-0.285	
session3:HF_typeHF_8	-0.002	-0.157	-0.156	-0.156	-0.158	-0.312	
session4:HF_typeHF_8	-0.002	-0.176	-0.174	-0.174	-0.176	-0.349	
activitystanding:HF_typeHF_2	0.001	-0.171	-0.086	-0.086	-0.087	-0.086	
activitywalking:HF_typeHF_2	-0.001	-0.234	-0.118	-0.117	-0.119	-0.119	
activitystanding:HF_typeHF_3	0.001	-0.086	-0.171	-0.085	-0.086	-0.086	
activitywalking:HF_typeHF_3	-0.001	-0.118	-0.234	-0.117	-0.118	-0.118	
activitystanding:HF_typeHF_5	0.001	-0.085	-0.085	-0.171	-0.086	-0.085	
activitywalking:HF_typeHF_5	-0.001	-0.117	-0.116	-0.234	-0.118	-0.117	
activitystanding:HF_typeHF_7	0.001	-0.087	-0.086	-0.086	-0.170	-0.087	
activitywalking:HF_typeHF_7	-0.001	-0.119	-0.118	-0.118	-0.233	-0.119	
activitystanding:HF_typeHF_8	0.001	-0.086	-0.085	-0.085	-0.086	-0.171	
activitywalking:HF_typeHF_8	-0.001	-0.118	-0.117	-0.117	-0.118	-0.235	
sexM:HF_typeHF_2	-0.692	-0.606	-0.300	-0.299	-0.304	-0.303	
sexM:HF_typeHF_3	-0.690	-0.302	-0.608	-0.299	-0.303	-0.302	
sexM:HF_typeHF_5	-0.688	-0.301	-0.299	-0.601	-0.302	-0.301	
sexM:HF_typeHF_7	-0.694	-0.304	-0.301	-0.301	-0.611	-0.304	
sexM:HF_typeHF_8	-0.692	-0.303	-0.300	-0.299	-0.304	-0.606	
	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7			
day2							
session2							
session3							
session4							

activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3	0.498			
day2:HF_typeHF_5	0.494	0.488		
day2:HF_typeHF_7	0.506	0.500	0.496	
day2:HF_typeHF_8	0.503	0.498	0.493	0.506
session2:HF_typeHF_2	0.033	0.016	0.015	0.016
session3:HF_typeHF_2	-0.002	-0.002	-0.002	-0.003
session4:HF_typeHF_2	-0.009	-0.006	-0.006	-0.006
session2:HF_typeHF_3	0.016	0.033	0.015	0.016
session3:HF_typeHF_3	-0.002	-0.001	-0.002	-0.003
session4:HF_typeHF_3	-0.006	-0.006	-0.006	-0.006
session2:HF_typeHF_5	0.016	0.016	0.033	0.016
session3:HF_typeHF_5	-0.002	-0.002	-0.002	-0.003
session4:HF_typeHF_5	-0.006	-0.006	-0.009	-0.006
session2:HF_typeHF_7	0.016	0.016	0.016	0.034
session3:HF_typeHF_7	-0.003	-0.002	-0.002	-0.002
session4:HF_typeHF_7	-0.006	-0.006	-0.006	-0.009
session2:HF_typeHF_8	0.016	0.016	0.015	0.016
session3:HF_typeHF_8	-0.003	-0.002	-0.002	-0.003
session4:HF_typeHF_8	-0.006	-0.006	-0.006	-0.006
activitystanding:HF_typeHF_2	0.018	0.009	0.009	0.010
activitywalking:HF_typeHF_2	0.007	0.003	0.003	0.003
activitystanding:HF_typeHF_3	0.009	0.019	0.009	0.010
activitywalking:HF_typeHF_3	0.003	0.008	0.002	0.003
activitystanding:HF_typeHF_5	0.009	0.009	0.018	0.010
activitywalking:HF_typeHF_5	0.003	0.003	0.007	0.003
activitystanding:HF_typeHF_7	0.010	0.009	0.009	0.017
activitywalking:HF_typeHF_7	0.003	0.003	0.003	0.007
activitystanding:HF_typeHF_8	0.009	0.009	0.009	0.010
activitywalking:HF_typeHF_8	0.003	0.003	0.003	0.003
sexM:HF_typeHF_2	-0.021	-0.012	-0.012	-0.011
sexM:HF_typeHF_3	-0.012	-0.010	-0.012	-0.011
sexM:HF_typeHF_5	-0.012	-0.012	-0.021	-0.011
sexM:HF_typeHF_7	-0.011	-0.011	-0.011	-0.011
sexM:HF_typeHF_8	-0.012	-0.012	-0.012	-0.011
	d2:HF_HF_8	s2:HF_HF_2	s3:HF_HF_2	s4:HF_HF_2
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				

HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2	0.016			
session3:HF_typeHF_2	-0.003	0.582		
session4:HF_typeHF_2	-0.006	0.528	0.665	
session2:HF_typeHF_3	0.016	0.504	0.290	0.264
session3:HF_typeHF_3	-0.002	0.290	0.500	0.332
session4:HF_typeHF_3	-0.006	0.264	0.332	0.498
session2:HF_typeHF_5	0.016	0.501	0.288	0.263
session3:HF_typeHF_5	-0.002	0.288	0.497	0.331
session4:HF_typeHF_5	-0.006	0.263	0.331	0.497
session2:HF_typeHF_7	0.016	0.508	0.292	0.266
session3:HF_typeHF_7	-0.003	0.293	0.505	0.336
session4:HF_typeHF_7	-0.006	0.267	0.336	0.505
session2:HF_typeHF_8	0.034	0.506	0.291	0.265
session3:HF_typeHF_8	-0.003	0.291	0.502	0.334
session4:HF_typeHF_8	-0.009	0.266	0.334	0.502
activitystanding:HF_typeHF_2	0.009	0.090	0.083	0.132
activitywalking:HF_typeHF_2	0.003	0.394	0.158	0.449
activitystanding:HF_typeHF_3	0.009	0.046	0.039	0.065
activitywalking:HF_typeHF_3	0.003	0.198	0.078	0.225
activitystanding:HF_typeHF_5	0.009	0.045	0.039	0.065
activitywalking:HF_typeHF_5	0.003	0.197	0.077	0.224
activitystanding:HF_typeHF_7	0.010	0.046	0.039	0.066
activitywalking:HF_typeHF_7	0.003	0.200	0.079	0.227
activitystanding:HF_typeHF_8	0.018	0.046	0.039	0.065
activitywalking:HF_typeHF_8	0.009	0.199	0.078	0.225
sexM:HF_typeHF_2	-0.012	0.002	0.001	0.002
sexM:HF_typeHF_3	-0.012	0.002	0.001	0.002
sexM:HF_typeHF_5	-0.012	0.002	0.001	0.002
sexM:HF_typeHF_7	-0.011	0.002	0.001	0.002
sexM:HF_typeHF_8	-0.021	0.002	0.001	0.002
	s2:HF_HF_3	s3:HF_HF_3	s4:HF_HF_3	s2:HF_HF_5
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				

day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3	0.582			
session4:HF_typeHF_3	0.527	0.663		
session2:HF_typeHF_5	0.498	0.286	0.261	
session3:HF_typeHF_5	0.286	0.495	0.328	0.581
session4:HF_typeHF_5	0.261	0.329	0.493	0.528
session2:HF_typeHF_7	0.505	0.290	0.264	0.502
session3:HF_typeHF_7	0.291	0.502	0.333	0.289
session4:HF_typeHF_7	0.266	0.334	0.501	0.264
session2:HF_typeHF_8	0.503	0.289	0.263	0.500
session3:HF_typeHF_8	0.289	0.499	0.331	0.287
session4:HF_typeHF_8	0.264	0.332	0.498	0.263
activitystanding:HF_typeHF_2	0.046	0.039	0.065	0.046
activitywalking:HF_typeHF_2	0.198	0.078	0.224	0.197
activitystanding:HF_typeHF_3	0.091	0.084	0.132	0.045
activitywalking:HF_typeHF_3	0.395	0.159	0.448	0.196
activitystanding:HF_typeHF_5	0.045	0.038	0.064	0.092
activitywalking:HF_typeHF_5	0.196	0.077	0.222	0.395
activitystanding:HF_typeHF_7	0.046	0.039	0.065	0.046
activitywalking:HF_typeHF_7	0.199	0.078	0.225	0.198
activitystanding:HF_typeHF_8	0.046	0.039	0.065	0.045
activitywalking:HF_typeHF_8	0.198	0.078	0.224	0.197
sexM:HF_typeHF_2	0.002	0.001	0.002	0.002
sexM:HF_typeHF_3	0.002	0.002	0.003	0.002
sexM:HF_typeHF_5	0.002	0.001	0.002	0.002
sexM:HF_typeHF_7	0.002	0.001	0.002	0.002
sexM:HF_typeHF_8	0.002	0.001	0.002	0.002
	s3:HF_HF_5	s4:HF_HF_5	s2:HF_HF_7	s3:HF_HF_7
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				

session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.665			
session2:HF_typeHF_7	0.289	0.263		
session3:HF_typeHF_7	0.500	0.332	0.579	
session4:HF_typeHF_7	0.332	0.499	0.526	0.664
session2:HF_typeHF_8	0.288	0.263	0.507	0.292
session3:HF_typeHF_8	0.496	0.330	0.291	0.504
session4:HF_typeHF_8	0.331	0.497	0.266	0.336
activitystanding:HF_typeHF_2	0.039	0.065	0.046	0.039
activitywalking:HF_typeHF_2	0.078	0.224	0.200	0.079
activitystanding:HF_typeHF_3	0.038	0.064	0.046	0.039
activitywalking:HF_typeHF_3	0.077	0.222	0.199	0.078
activitystanding:HF_typeHF_5	0.082	0.132	0.046	0.039
activitywalking:HF_typeHF_5	0.158	0.449	0.198	0.078
activitystanding:HF_typeHF_7	0.039	0.065	0.090	0.083
activitywalking:HF_typeHF_7	0.078	0.225	0.393	0.158
activitystanding:HF_typeHF_8	0.038	0.065	0.046	0.039
activitywalking:HF_typeHF_8	0.077	0.223	0.199	0.079
sexM:HF_typeHF_2	0.001	0.002	0.002	0.001
sexM:HF_typeHF_3	0.001	0.002	0.002	0.001
sexM:HF_typeHF_5	0.001	0.002	0.002	0.001
sexM:HF_typeHF_7	0.001	0.002	0.001	0.002
sexM:HF_typeHF_8	0.001	0.002	0.002	0.001
	s4:HF_HF_7	s2:HF_HF_8	s3:HF_HF_8	s4:HF_HF_8
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				

session4:HF_typeHF_7				
session2:HF_typeHF_8	0.267			
session3:HF_typeHF_8	0.335	0.577		
session4:HF_typeHF_8	0.505	0.528	0.664	
activitystanding:HF_typeHF_2	0.066	0.046	0.039	0.066
activitywalking:HF_typeHF_2	0.227	0.199	0.078	0.226
activitystanding:HF_typeHF_3	0.066	0.046	0.039	0.065
activitywalking:HF_typeHF_3	0.226	0.198	0.078	0.225
activitystanding:HF_typeHF_5	0.065	0.045	0.038	0.065
activitywalking:HF_typeHF_5	0.225	0.197	0.077	0.224
activitystanding:HF_typeHF_7	0.131	0.046	0.039	0.066
activitywalking:HF_typeHF_7	0.448	0.200	0.079	0.227
activitystanding:HF_typeHF_8	0.066	0.092	0.079	0.130
activitywalking:HF_typeHF_8	0.227	0.396	0.153	0.447
sexM:HF_typeHF_2	0.002	0.002	0.001	0.002
sexM:HF_typeHF_3	0.002	0.002	0.001	0.002
sexM:HF_typeHF_5	0.002	0.002	0.001	0.002
sexM:HF_typeHF_7	0.001	0.002	0.001	0.002
sexM:HF_typeHF_8	0.002	0.002	0.001	0.001
		actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2	0.401			

activitystanding:HF_typeHF_3	0.505	0.203	
activitywalking:HF_typeHF_3	0.204	0.502	0.403
activitystanding:HF_typeHF_5	0.503	0.202	0.499
activitywalking:HF_typeHF_5	0.203	0.499	0.201
activitystanding:HF_typeHF_7	0.511	0.206	0.507
activitywalking:HF_typeHF_7	0.206	0.507	0.204
activitystanding:HF_typeHF_8	0.507	0.204	0.503
activitywalking:HF_typeHF_8	0.204	0.503	0.203
sexM:HF_typeHF_2	0.000	0.001	0.000
sexM:HF_typeHF_3	0.000	0.001	0.000
sexM:HF_typeHF_5	0.000	0.001	0.000
sexM:HF_typeHF_7	-0.001	0.001	-0.001
sexM:HF_typeHF_8	0.000	0.001	0.000
	actvtyw:HF_HF_3	actvtys:HF_HF_5	actvtyw:HF_HF_5
day2			
session2			
session3			
session4			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
session2:HF_typeHF_2			
session3:HF_typeHF_2			
session4:HF_typeHF_2			
session2:HF_typeHF_3			
session3:HF_typeHF_3			
session4:HF_typeHF_3			
session2:HF_typeHF_5			
session3:HF_typeHF_5			
session4:HF_typeHF_5			
session2:HF_typeHF_7			
session3:HF_typeHF_7			
session4:HF_typeHF_7			
session2:HF_typeHF_8			
session3:HF_typeHF_8			
session4:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5	0.201		
activitywalking:HF_typeHF_5	0.496	0.402	
activitystanding:HF_typeHF_7	0.204	0.505	0.203
activitywalking:HF_typeHF_7	0.504	0.203	0.502

activitystanding:HF_typeHF_8	0.203	0.500	0.201
activitywalking:HF_typeHF_8	0.500	0.202	0.498
sexM:HF_typeHF_2	0.001	0.000	0.001
sexM:HF_typeHF_3	0.002	0.000	0.001
sexM:HF_typeHF_5	0.001	0.000	0.001
sexM:HF_typeHF_7	0.001	-0.001	0.001
sexM:HF_typeHF_8	0.001	0.000	0.001
	actvtys:HF_HF_7	actvtyw:HF_HF_7	actvtys:HF_HF_8
day2			
session2			
session3			
session4			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
session2:HF_typeHF_2			
session3:HF_typeHF_2			
session4:HF_typeHF_2			
session2:HF_typeHF_3			
session3:HF_typeHF_3			
session4:HF_typeHF_3			
session2:HF_typeHF_5			
session3:HF_typeHF_5			
session4:HF_typeHF_5			
session2:HF_typeHF_7			
session3:HF_typeHF_7			
session4:HF_typeHF_7			
session2:HF_typeHF_8			
session3:HF_typeHF_8			
session4:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5			
activitywalking:HF_typeHF_5			
activitystanding:HF_typeHF_7			
activitywalking:HF_typeHF_7	0.402		
activitystanding:HF_typeHF_8	0.508	0.205	
activitywalking:HF_typeHF_8	0.205	0.506	0.402
sexM:HF_typeHF_2	0.000	0.001	0.000
sexM:HF_typeHF_3	0.000	0.001	0.000
sexM:HF_typeHF_5	0.000	0.001	0.000
sexM:HF_typeHF_7	-0.001	0.001	-0.001

sexM:HF_typeHF_8	0.000	0.001	0.000	
	actvtyw:HF_HF_8	sM:HF_HF_2	sM:HF_HF_3	sM:HF_HF_5
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				
activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2	0.001			
sexM:HF_typeHF_3	0.001	0.500		
sexM:HF_typeHF_5	0.001	0.498	0.497	
sexM:HF_typeHF_7	0.001	0.502	0.500	0.499
sexM:HF_typeHF_8	0.002	0.501	0.500	0.498
	sM:HF_HF_7			
day2				
session2				
session3				
session4				

```

activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5
activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.501

Standardized Within-Group Residuals:
      Min      Q1      Med      Q3      Max
-5.67505785 -0.38532295  0.09131612  0.54637192  3.54032939

Number of Observations: 2638
Number of Groups:
      person HF_type %in% person
      24          144

```

```

> mdl7_reml <- m3_ar1
> anova_md17 <- anova(mdl6_reml, mdl7_reml)
> anova_md17_clean <- data.frame(
+   Model = c("Assumes Compound Symmetry", "Assumes Auto-regressive AR(1)"),
+   Df = as.numeric(anova_md17$df),
+   AIC = round(as.numeric(anova_md17$AIC)),
+   BIC = round(as.numeric(anova_md17$BIC)),
+   LogLik = round(as.numeric(anova_md17$logLik)),
+   LRT_Stat = c(NA, round(anova_md17$L.Ratio[2], 2)),
+   p_value = c(NA, anova_md17$p-value[2]))
>
> print(anova_md17)

```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl6_reml		1 53	311.9480	622.4968	-102.9740			
mdl7_reml		2 54	-402.5175	-86.1092	255.2587	1 vs 2	716.4654	<.0001

```

> anova_md17_clean <- anova_md17_clean %>%
+   mutate(
+     LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
+     p_value = ifelse(is.na(p_value), "--",
+       # eps defines the threshold for "<"
+       # digits defines the rounding for values above that threshold
+       format.pval(p_value, eps = 0.0001, digits = 4))
+   )
>
>

```

```

> col_labels <- c("Model Version",
+   "\\shortstack{Degrees of\\\\Freedom}",
+   "AIC",
+   "BIC",
+   "Log-Likelihood",
+   "\\shortstack{Likelihood Ratio\\\\Test Stat}",
+   "P-Value")

```

```

> kable(anova_md17_clean,
+   format="latex",
+   col.names = col_labels,
+   align = "lrrrrrr",
+   booktabs = TRUE, #, #right aligns numbers
+   escape = FALSE,
+   caption = "\\footnotesize{Likelihood test between the model assuming compound symmetry and AR1,
+   kable_styling(latex_options = c("scale_down","hold_position"),
+   full_width = FALSE,
+   table.envir = "table") %>%
+   add_header_above(c("Likelihood Ratio Test for Covariance Structures" = 7),
+   bold = TRUE, align = "l") %>%
+   row_spec(0, bold = TRUE)

```

5.5 Covariance Structure (ARH(1))

We saw there is a temporal aspect to the data, where consecutive sessions are more closely linked than further ones. However, AR(1) assumes that the amount of variance is identical for all sensors. However, we would expect sensors on different body parts to have different variance. As such, we tested a heterogeneous AR

Table 3: Likelihood test between the model assuming compound symmetry and AR1, testing an AR1 structure where magnitude of correlation depends on amount of time between observations (instead of flat correlation). The AR1 model has significantly lower AIC and BIC and a very small p-value from the likelihood ratio test (it is statistically significant). Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Assumes Compound Symmetry	53	312	622	-103	–	–
Assumes Auto-regressive AR(1)	54	-403	-86	255	716.47	< 1e-04

structure ARH(1) by combining the AR(1) structure with heterogeneous residual variances across sensors.

The ρ from this model is 0.6331, slightly higher than the previous model's (0.5814), suggesting the temporal aspect of the data was higher than previously suggested and was masked by the noise levels of the sensors. The estimates for the heterogenous variances relative to sensor 16 (baseline, dominant hand) are: 1.27 (sensor 2, chest (side)), 0.73 (sensor 3, upper chest), 0.35 (sensor 5, upper arm), 0.95 (sensor 7, lower arm), and 0.97 (sensor 8, abdomen). This implies that sensors on the abdomen, lower arm, and dominant hand have similar noise. The sensor on the chest (side) is significantly noisier than those sensors, and the sensors on the upper chest and arm are significantly less noisy. There does not seem to be an obvious trend or reason for this; it may be helpful to consult experts in the field. Additionally, Table 4 shows that the ARH(1) structure has significantly lower AIC and BIC than the AR(1) structure, and is statistically significant in the likelihood ratio test.

```
> m3_arh1 <- lme(
+   fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "REML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
> anova(m3_ar1, m3_arh1)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
m3_ar1	1	54	-402.5175	-86.1092	255.2587			
m3_arh1	2	59	-944.8613	-599.1559	531.4307	1 vs 2	552.3438	<.0001

```
> mdl8_reml <- m3_arh1
> anova_md18 <- anova(mdl7_reml, mdl8_reml)
>
>
> anova_md18_clean <- data.frame(
+   Model = c("Assumes Auto-regressive Structure (AR(1))", "Assumes Heterogenous AR Structure (ARH(1))"),
+   Df = as.numeric(anova_md18$df),
+   AIC = round(as.numeric(anova_md18$AIC)),
+   BIC = round(as.numeric(anova_md18$BIC)),
+   LogLik = round(as.numeric(anova_md18$logLik)),
+   LRT_Stat = c(NA, round(anova_md18$L.Ratio[2], 2)),
+   p_value = c(NA, anova_md18$p-value[2]))
```

```

>
>
> anova_md18_clean <- anova_md18_clean %>%
+   mutate(
+     LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
+     p_value = ifelse(is.na(p_value), "--",
+                       # eps defines the threshold for "<"
+                       # digits defines the rounding for values above that threshold
+                       format.pval(p_value, eps = 0.0001, digits = 4))
+   )
>
> col_labels <- c("Model Version",
+                 "\\shortstack{Degrees of\\\\Freedom}",
+                 "AIC",
+                 "BIC",
+                 "Log-Likelihood",
+                 "\\shortstack{Likelihood Ratio\\\\Test Stat}",
+                 "P-Value")
>
> kable(anova_md18_clean,
+       format="latex",
+       col.names = col_labels,
+       align = "lrrrrrr",
+       booktabs = TRUE, #, #right aligns numbers
+       escape = FALSE,
+       caption = "\\footnotesize{Likelihood test between AR1 model and ARH1 model, testing an ARH1 str",
+       kable_styling(latex_options = c("scale_down", "hold_position"),
+                     full_width = FALSE,
+                     table.envir = "table") %>%
+       add_header_above(c("Likelihood Ratio Test for Covariance Structures" = 7),
+                         bold = TRUE, align = "l") %>%
+       row_spec(0, bold = TRUE)

```

Table 4: Likelihood test between AR1 model and ARH1 model, testing an ARH1 structure that allows each sensor to have its own unique residual variance instead of assuming uniform residual noise. Model 8 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Assumes Auto-regressive Structure (AR(1))	54	-403	-86	255	—	—
Assumes Heterogeneous AR Structure (ARH(1))	59	-945	-599	531	552.34	< 1e-04

5.6 Prune Fixed Effects

```
> summary(m3_arh1)
```

Linear mixed-effects model fit by REML

Data: data_clean

```

      AIC      BIC   logLik
-944.8613 -599.1559 531.4307

```

Random effects:

```

Formula: ~1 + day | person
Structure: General positive-definite, Log-Cholesky parametrization
          StdDev      Corr
(Intercept) 0.08188245 (Intr)
day2         0.13861289 -0.914

Formula: ~1 | HF_type %in% person
          (Intercept) Residual
StdDev:    0.1604554 0.2777172

Correlation Structure: ARMA(1,0)
Formula: ~time_step | person/HF_type/day
Parameter estimate(s):
      Phi1
0.6331363
Variance function:
Structure: Different standard deviations per stratum
Formula: ~1 | HF_type
Parameter estimates:
      HF_16      HF_2      HF_3      HF_5      HF_7      HF_8
1.0000000 1.2728633 0.7262636 0.3513202 0.9468251 0.9695869
Fixed effects: log(HF_value) ~ (day + session + activity + sex) * HF_type
              Value Std.Error   DF  t-value p-value
(Intercept)      4.493373 0.07042583 2458  63.80291 0.0000
day2             -0.143927 0.05571199 2458  -2.58340 0.0098
session2         -0.062673 0.03645385 2458  -1.71925 0.0857
session3         -0.097691 0.04005609 2458  -2.43887 0.0148
session4         -0.231815 0.05027809 2458  -4.61065 0.0000
activitystanding -0.090755 0.01727363 2458  -5.25395 0.0000
activitywalking   0.036713 0.02551112 2458   1.43910 0.1503
sexM             -0.003146 0.08232297   22  -0.03821 0.9699
HF_typeHF_2      -0.721244 0.10380919  110  -6.94779 0.0000
HF_typeHF_3      -0.635539 0.09062000  110  -7.01324 0.0000
HF_typeHF_5      -0.428941 0.08455433  110  -5.07296 0.0000
HF_typeHF_7      -0.172980 0.09466133  110  -1.82736 0.0704
HF_typeHF_8      -0.918911 0.09543332  110  -9.62883 0.0000
day2:HF_typeHF_2   0.467404 0.07753128 2458   6.02859 0.0000
day2:HF_typeHF_3   0.462812 0.05984338 2458   7.73373 0.0000
day2:HF_typeHF_5   0.410355 0.05151834 2458   7.96522 0.0000
day2:HF_typeHF_7   0.304886 0.06550692 2458   4.65425 0.0000
day2:HF_typeHF_8   0.460062 0.06685170 2458   6.88183 0.0000
session2:HF_typeHF_2 0.102077 0.05855151 2458   1.74337 0.0814
session3:HF_typeHF_2 0.117142 0.06461886 2458   1.81281 0.0700
session4:HF_typeHF_2 0.315861 0.08117610 2458   3.89106 0.0001
session2:HF_typeHF_3 0.036208 0.04503545 2458   0.80398 0.4215
session3:HF_typeHF_3 0.080630 0.04960008 2458   1.62560 0.1042
session4:HF_typeHF_3 0.215550 0.06237625 2458   3.45564 0.0006
session2:HF_typeHF_5 0.071107 0.03868081 2458   1.83829 0.0661
session3:HF_typeHF_5 0.091349 0.04253096 2458   2.14783 0.0318
session4:HF_typeHF_5 0.233022 0.05339421 2458   4.36418 0.0000
session2:HF_typeHF_7 0.035111 0.04979933 2458   0.70505 0.4808
session3:HF_typeHF_7 0.040911 0.05477687 2458   0.74687 0.4552
session4:HF_typeHF_7 0.107164 0.06878349 2458   1.55799 0.1194
session2:HF_typeHF_8 0.071478 0.05052978 2458   1.41458 0.1573

```

session3:HF_typeHF_8	0.158104	0.05574327	2458	2.83628	0.0046		
session4:HF_typeHF_8	0.272448	0.06990019	2458	3.89767	0.0001		
activitystanding:HF_typeHF_2	-0.088205	0.02763670	2458	-3.19157	0.0014		
activitywalking:HF_typeHF_2	-0.375987	0.04103149	2458	-9.16338	0.0000		
activitystanding:HF_typeHF_3	0.190141	0.02132557	2458	8.91609	0.0000		
activitywalking:HF_typeHF_3	0.221246	0.03155348	2458	7.01179	0.0000		
activitystanding:HF_typeHF_5	0.064581	0.01832264	2458	3.52466	0.0004		
activitywalking:HF_typeHF_5	-0.048935	0.02707821	2458	-1.80716	0.0709		
activitystanding:HF_typeHF_7	-0.032102	0.02350491	2458	-1.36576	0.1721		
activitywalking:HF_typeHF_7	-0.116329	0.03481419	2458	-3.34141	0.0008		
activitystanding:HF_typeHF_8	0.177035	0.02396926	2458	7.38590	0.0000		
activitywalking:HF_typeHF_8	0.146171	0.03546881	2458	4.12111	0.0000		
sexM:HF_typeHF_2	-0.073045	0.12078038	110	-0.60477	0.5466		
sexM:HF_typeHF_3	0.069210	0.11026672	110	0.62766	0.5315		
sexM:HF_typeHF_5	0.019658	0.10596852	110	0.18551	0.8532		
sexM:HF_typeHF_7	0.136107	0.11341981	110	1.20003	0.2327		
sexM:HF_typeHF_8	0.093164	0.11422595	110	0.81561	0.4165		
Correlation:							
	(Intr)	day2	sessn2	sessn3	sessn4	actvtys	actvtyw
day2	-0.391						
session2	-0.297	0.022					
session3	-0.332	-0.004	0.604				
session4	-0.372	-0.010	0.551	0.694			
activitystanding	-0.180	0.013	0.106	0.101	0.160		
activitywalking	-0.250	0.003	0.414	0.193	0.476	0.414	
sexM	-0.572	-0.019	0.003	0.003	0.003	-0.001	0.003
HF_typeHF_2	-0.634	0.190	0.202	0.225	0.252	0.122	0.169
HF_typeHF_3	-0.726	0.218	0.231	0.258	0.289	0.140	0.194
HF_typeHF_5	-0.778	0.234	0.248	0.277	0.310	0.150	0.208
HF_typeHF_7	-0.695	0.209	0.221	0.247	0.277	0.134	0.186
HF_typeHF_8	-0.689	0.207	0.219	0.245	0.275	0.133	0.184
day2:HF_typeHF_2	0.201	-0.533	-0.016	0.003	0.007	-0.009	-0.002
day2:HF_typeHF_3	0.261	-0.691	-0.021	0.003	0.009	-0.012	-0.003
day2:HF_typeHF_5	0.303	-0.803	-0.024	0.004	0.011	-0.014	-0.003
day2:HF_typeHF_7	0.239	-0.631	-0.019	0.003	0.008	-0.011	-0.002
day2:HF_typeHF_8	0.234	-0.619	-0.018	0.003	0.008	-0.011	-0.002
session2:HF_typeHF_2	0.185	-0.014	-0.623	-0.376	-0.343	-0.066	-0.258
session3:HF_typeHF_2	0.206	0.002	-0.374	-0.620	-0.430	-0.063	-0.120
session4:HF_typeHF_2	0.231	0.006	-0.341	-0.430	-0.619	-0.099	-0.295
session2:HF_typeHF_3	0.241	-0.018	-0.809	-0.489	-0.446	-0.086	-0.335
session3:HF_typeHF_3	0.268	0.003	-0.488	-0.808	-0.560	-0.082	-0.156
session4:HF_typeHF_3	0.300	0.008	-0.444	-0.559	-0.806	-0.129	-0.384
session2:HF_typeHF_5	0.280	-0.021	-0.942	-0.569	-0.520	-0.100	-0.390
session3:HF_typeHF_5	0.313	0.004	-0.569	-0.942	-0.653	-0.095	-0.182
session4:HF_typeHF_5	0.350	0.009	-0.519	-0.653	-0.942	-0.150	-0.448
session2:HF_typeHF_7	0.218	-0.016	-0.732	-0.442	-0.404	-0.077	-0.303
session3:HF_typeHF_7	0.243	0.003	-0.441	-0.731	-0.507	-0.074	-0.141
session4:HF_typeHF_7	0.272	0.007	-0.403	-0.507	-0.731	-0.117	-0.348
session2:HF_typeHF_8	0.215	-0.016	-0.721	-0.435	-0.398	-0.076	-0.299
session3:HF_typeHF_8	0.239	0.003	-0.434	-0.719	-0.499	-0.073	-0.139
session4:HF_typeHF_8	0.268	0.007	-0.397	-0.499	-0.719	-0.115	-0.343
activitystanding:HF_typeHF_2	0.113	-0.008	-0.066	-0.063	-0.100	-0.625	-0.259
activitywalking:HF_typeHF_2	0.155	-0.002	-0.257	-0.120	-0.296	-0.257	-0.622
activitystanding:HF_typeHF_3	0.146	-0.010	-0.086	-0.082	-0.129	-0.810	-0.335

activitywalking:HF_typeHF_3	0.202	-0.002	-0.335	-0.156	-0.385	-0.335	-0.809
activitystanding:HF_typeHF_5	0.170	-0.012	-0.100	-0.095	-0.150	-0.943	-0.390
activitywalking:HF_typeHF_5	0.235	-0.003	-0.390	-0.182	-0.449	-0.390	-0.942
activitystanding:HF_typeHF_7	0.132	-0.009	-0.078	-0.074	-0.117	-0.735	-0.304
activitywalking:HF_typeHF_7	0.183	-0.002	-0.303	-0.142	-0.349	-0.303	-0.733
activitystanding:HF_typeHF_8	0.130	-0.009	-0.076	-0.073	-0.115	-0.721	-0.298
activitywalking:HF_typeHF_8	0.180	-0.002	-0.298	-0.139	-0.342	-0.298	-0.719
sexM:HF_typeHF_2	0.379	0.013	-0.002	-0.002	-0.002	0.000	-0.002
sexM:HF_typeHF_3	0.415	0.014	-0.003	-0.002	-0.003	0.000	-0.002
sexM:HF_typeHF_5	0.432	0.015	-0.003	-0.002	-0.003	0.000	-0.002
sexM:HF_typeHF_7	0.404	0.014	-0.002	-0.002	-0.002	0.001	-0.002
sexM:HF_typeHF_8	0.401	0.014	-0.002	-0.002	-0.002	0.000	-0.002
sexM	HF_HF_2	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8		
day2							
session2							
session3							
session4							
activitystanding							
activitywalking							
sexM							
HF_typeHF_2	0.377						
HF_typeHF_3	0.432	0.493					
HF_typeHF_5	0.463	0.528	0.605				
HF_typeHF_7	0.414	0.472	0.540	0.579			
HF_typeHF_8	0.410	0.468	0.536	0.575	0.513		
day2:HF_typeHF_2	0.014	-0.360	-0.157	-0.169	-0.150	-0.149	
day2:HF_typeHF_3	0.018	-0.178	-0.327	-0.219	-0.194	-0.194	
day2:HF_typeHF_5	0.021	-0.207	-0.237	-0.297	-0.226	-0.225	
day2:HF_typeHF_7	0.016	-0.162	-0.185	-0.199	-0.333	-0.176	
day2:HF_typeHF_8	0.016	-0.159	-0.182	-0.196	-0.174	-0.337	
session2:HF_typeHF_2	-0.002	-0.325	-0.144	-0.154	-0.138	-0.137	
session3:HF_typeHF_2	-0.002	-0.364	-0.160	-0.172	-0.153	-0.152	
session4:HF_typeHF_2	-0.002	-0.406	-0.179	-0.192	-0.171	-0.170	
session2:HF_typeHF_3	-0.003	-0.163	-0.286	-0.200	-0.179	-0.178	
session3:HF_typeHF_3	-0.002	-0.182	-0.320	-0.223	-0.200	-0.198	
session4:HF_typeHF_3	-0.003	-0.203	-0.359	-0.250	-0.223	-0.221	
session2:HF_typeHF_5	-0.003	-0.190	-0.218	-0.263	-0.209	-0.207	
session3:HF_typeHF_5	-0.003	-0.212	-0.243	-0.294	-0.233	-0.231	
session4:HF_typeHF_5	-0.003	-0.238	-0.273	-0.329	-0.261	-0.259	
session2:HF_typeHF_7	-0.002	-0.148	-0.169	-0.181	-0.301	-0.161	
session3:HF_typeHF_7	-0.002	-0.165	-0.189	-0.202	-0.339	-0.179	
session4:HF_typeHF_7	-0.002	-0.185	-0.212	-0.227	-0.378	-0.201	
session2:HF_typeHF_8	-0.002	-0.146	-0.167	-0.179	-0.160	-0.306	
session3:HF_typeHF_8	-0.002	-0.162	-0.186	-0.199	-0.178	-0.339	
session4:HF_typeHF_8	-0.002	-0.182	-0.208	-0.223	-0.199	-0.381	
activitystanding:HF_typeHF_2	0.000	-0.197	-0.087	-0.094	-0.084	-0.083	
activitywalking:HF_typeHF_2	-0.002	-0.271	-0.121	-0.129	-0.116	-0.115	
activitystanding:HF_typeHF_3	0.001	-0.099	-0.174	-0.121	-0.108	-0.108	
activitywalking:HF_typeHF_3	-0.002	-0.137	-0.240	-0.168	-0.150	-0.149	
activitystanding:HF_typeHF_5	0.001	-0.115	-0.132	-0.159	-0.126	-0.125	
activitywalking:HF_typeHF_5	-0.002	-0.160	-0.183	-0.221	-0.175	-0.174	
activitystanding:HF_typeHF_7	0.001	-0.090	-0.103	-0.110	-0.183	-0.098	
activitywalking:HF_typeHF_7	-0.002	-0.124	-0.142	-0.152	-0.252	-0.135	
activitystanding:HF_typeHF_8	0.001	-0.088	-0.101	-0.108	-0.097	-0.185	

activitywalking:HF_typeHF_8	-0.002	-0.122	-0.140	-0.150	-0.134	-0.256
sexM:HF_typeHF_2	-0.663	-0.566	-0.294	-0.315	-0.282	-0.279
sexM:HF_typeHF_3	-0.726	-0.281	-0.601	-0.345	-0.309	-0.306
sexM:HF_typeHF_5	-0.756	-0.293	-0.335	-0.615	-0.321	-0.318
sexM:HF_typeHF_7	-0.706	-0.274	-0.314	-0.336	-0.592	-0.298
sexM:HF_typeHF_8	-0.701	-0.272	-0.311	-0.333	-0.298	-0.585
	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7		
day2						
session2						
session3						
session4						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5						
HF_typeHF_7						
HF_typeHF_8						
day2:HF_typeHF_2						
day2:HF_typeHF_3	0.499					
day2:HF_typeHF_5	0.580	0.752				
day2:HF_typeHF_7	0.454	0.588	0.683			
day2:HF_typeHF_8	0.447	0.579	0.673	0.526		
session2:HF_typeHF_2	0.027	0.013	0.015	0.012		
session3:HF_typeHF_2	-0.001	-0.002	-0.003	-0.002		
session4:HF_typeHF_2	-0.008	-0.005	-0.007	-0.005		
session2:HF_typeHF_3	0.013	0.027	0.019	0.015		
session3:HF_typeHF_3	-0.002	-0.002	-0.003	-0.003		
session4:HF_typeHF_3	-0.006	-0.007	-0.009	-0.007		
session2:HF_typeHF_5	0.015	0.019	0.025	0.018		
session3:HF_typeHF_5	-0.003	-0.003	-0.004	-0.003		
session4:HF_typeHF_5	-0.007	-0.008	-0.011	-0.008		
session2:HF_typeHF_7	0.012	0.015	0.017	0.027		
session3:HF_typeHF_7	-0.002	-0.003	-0.003	-0.002		
session4:HF_typeHF_7	-0.005	-0.006	-0.008	-0.008		
session2:HF_typeHF_8	0.011	0.015	0.017	0.014		
session3:HF_typeHF_8	-0.002	-0.002	-0.003	-0.002		
session4:HF_typeHF_8	-0.005	-0.006	-0.008	-0.006		
activitystanding:HF_typeHF_2	0.014	0.007	0.009	0.007		
activitywalking:HF_typeHF_2	0.006	0.002	0.002	0.002		
activitystanding:HF_typeHF_3	0.007	0.015	0.011	0.009		
activitywalking:HF_typeHF_3	0.002	0.005	0.002	0.002		
activitystanding:HF_typeHF_5	0.009	0.011	0.014	0.010		
activitywalking:HF_typeHF_5	0.002	0.003	0.004	0.002		
activitystanding:HF_typeHF_7	0.007	0.009	0.010	0.013		
activitywalking:HF_typeHF_7	0.001	0.002	0.002	0.005		
activitystanding:HF_typeHF_8	0.007	0.009	0.010	0.008		
activitywalking:HF_typeHF_8	0.001	0.002	0.002	0.002		
sexM:HF_typeHF_2	-0.022	-0.014	-0.016	-0.011		
sexM:HF_typeHF_3	-0.012	-0.014	-0.018	-0.012		
sexM:HF_typeHF_5	-0.012	-0.016	-0.022	-0.013		
sexM:HF_typeHF_7	-0.010	-0.013	-0.015	-0.012		
sexM:HF_typeHF_8	-0.011	-0.015	-0.017	-0.012		

```

d2:HF_HF_8 s2:HF_HF_2 s3:HF_HF_2 s4:HF_HF_2

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2      0.011
session3:HF_typeHF_2      -0.002      0.609
session4:HF_typeHF_2      -0.005      0.553      0.694
session2:HF_typeHF_3      0.015      0.504      0.303      0.276
session3:HF_typeHF_3      -0.003      0.304      0.501      0.347
session4:HF_typeHF_3      -0.007      0.277      0.347      0.499
session2:HF_typeHF_5      0.017      0.587      0.353      0.322
session3:HF_typeHF_5      -0.003      0.354      0.584      0.405
session4:HF_typeHF_5      -0.008      0.323      0.405      0.583
session2:HF_typeHF_7      0.013      0.456      0.274      0.250
session3:HF_typeHF_7      -0.002      0.275      0.453      0.314
session4:HF_typeHF_7      -0.006      0.251      0.314      0.453
session2:HF_typeHF_8      0.028      0.449      0.270      0.246
session3:HF_typeHF_8      -0.003      0.270      0.445      0.309
session4:HF_typeHF_8      -0.008      0.247      0.309      0.446
activitystanding:HF_typeHF_2 0.007      0.105      0.109      0.163
activitywalking:HF_typeHF_2 0.001      0.414      0.197      0.476
activitystanding:HF_typeHF_3 0.009      0.053      0.051      0.080
activitywalking:HF_typeHF_3 0.002      0.208      0.097      0.238
activitystanding:HF_typeHF_5 0.010      0.062      0.059      0.093
activitywalking:HF_typeHF_5 0.002      0.243      0.113      0.278
activitystanding:HF_typeHF_7 0.008      0.048      0.046      0.073
activitywalking:HF_typeHF_7 0.002      0.189      0.088      0.216
activitystanding:HF_typeHF_8 0.015      0.047      0.045      0.071
activitywalking:HF_typeHF_8 0.007      0.185      0.086      0.212
sexM:HF_typeHF_2      -0.012      0.001      0.001      0.001
sexM:HF_typeHF_3      -0.014      0.002      0.001      0.002
sexM:HF_typeHF_5      -0.014      0.002      0.001      0.002
sexM:HF_typeHF_7      -0.012      0.002      0.001      0.001
sexM:HF_typeHF_8      -0.022      0.002      0.001      0.001
s2:HF_HF_3 s3:HF_HF_3 s4:HF_HF_3 s2:HF_HF_5

day2
session2
session3
session4
activitystanding

```

```

activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3      0.607
session4:HF_typeHF_3      0.552      0.693
session2:HF_typeHF_5      0.763      0.459      0.419
session3:HF_typeHF_5      0.460      0.761      0.527      0.605
session4:HF_typeHF_5      0.420      0.528      0.759      0.552
session2:HF_typeHF_7      0.593      0.357      0.325      0.690
session3:HF_typeHF_7      0.357      0.591      0.409      0.416
session4:HF_typeHF_7      0.326      0.410      0.589      0.380
session2:HF_typeHF_8      0.584      0.352      0.321      0.680
session3:HF_typeHF_8      0.351      0.580      0.402      0.409
session4:HF_typeHF_8      0.321      0.403      0.580      0.374
activitystanding:HF_typeHF_2 0.054      0.051      0.080      0.062
activitywalking:HF_typeHF_2 0.208      0.097      0.239      0.243
activitystanding:HF_typeHF_3 0.107      0.106      0.161      0.081
activitywalking:HF_typeHF_3 0.414      0.196      0.476      0.316
activitystanding:HF_typeHF_5 0.081      0.077      0.121      0.106
activitywalking:HF_typeHF_5 0.316      0.147      0.362      0.414
activitystanding:HF_typeHF_7 0.063      0.060      0.094      0.073
activitywalking:HF_typeHF_7 0.246      0.114      0.281      0.286
activitystanding:HF_typeHF_8 0.062      0.059      0.093      0.072
activitywalking:HF_typeHF_8 0.241      0.112      0.276      0.281
sexM:HF_typeHF_2      0.002      0.002      0.002      0.002
sexM:HF_typeHF_3      0.002      0.002      0.003      0.003
sexM:HF_typeHF_5      0.002      0.002      0.002      0.003
sexM:HF_typeHF_7      0.002      0.002      0.002      0.002
sexM:HF_typeHF_8      0.002      0.002      0.002      0.002
s3:HF_HF_5 s4:HF_HF_5 s2:HF_HF_7 s3:HF_HF_7
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7

```

HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5	0.694			
session2:HF_typeHF_7	0.416	0.380		
session3:HF_typeHF_7	0.689	0.478	0.605	
session4:HF_typeHF_7	0.478	0.688	0.551	0.693
session2:HF_typeHF_8	0.410	0.375	0.528	0.318
session3:HF_typeHF_8	0.677	0.469	0.318	0.525
session4:HF_typeHF_8	0.470	0.677	0.290	0.365
activitystanding:HF_typeHF_2	0.059	0.094	0.048	0.046
activitywalking:HF_typeHF_2	0.113	0.279	0.188	0.088
activitystanding:HF_typeHF_3	0.077	0.122	0.063	0.060
activitywalking:HF_typeHF_3	0.147	0.363	0.245	0.114
activitystanding:HF_typeHF_5	0.102	0.160	0.073	0.070
activitywalking:HF_typeHF_5	0.194	0.476	0.286	0.133
activitystanding:HF_typeHF_7	0.070	0.110	0.105	0.107
activitywalking:HF_typeHF_7	0.133	0.329	0.413	0.196
activitystanding:HF_typeHF_8	0.069	0.108	0.056	0.053
activitywalking:HF_typeHF_8	0.131	0.323	0.218	0.102
sexM:HF_typeHF_2	0.002	0.002	0.002	0.001
sexM:HF_typeHF_3	0.002	0.002	0.002	0.002
sexM:HF_typeHF_5	0.002	0.002	0.002	0.002
sexM:HF_typeHF_7	0.002	0.002	0.001	0.002
sexM:HF_typeHF_8	0.002	0.002	0.002	0.001
	s4:HF_HF_7	s2:HF_HF_8	s3:HF_HF_8	s4:HF_HF_8
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				

session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8	0.291			
session3:HF_typeHF_8	0.364	0.603		
session4:HF_typeHF_8	0.526	0.552	0.693	
activitystanding:HF_typeHF_2	0.073	0.048	0.045	0.072
activitywalking:HF_typeHF_2	0.216	0.186	0.086	0.213
activitystanding:HF_typeHF_3	0.094	0.062	0.059	0.093
activitywalking:HF_typeHF_3	0.281	0.242	0.112	0.277
activitystanding:HF_typeHF_5	0.110	0.072	0.068	0.108
activitywalking:HF_typeHF_5	0.328	0.281	0.131	0.323
activitystanding:HF_typeHF_7	0.160	0.056	0.053	0.084
activitywalking:HF_typeHF_7	0.475	0.219	0.102	0.251
activitystanding:HF_typeHF_8	0.084	0.107	0.102	0.160
activitywalking:HF_typeHF_8	0.250	0.416	0.191	0.475
sexM:HF_typeHF_2	0.002	0.002	0.001	0.002
sexM:HF_typeHF_3	0.002	0.002	0.002	0.002
sexM:HF_typeHF_5	0.002	0.002	0.002	0.002
sexM:HF_typeHF_7	0.002	0.002	0.001	0.002
sexM:HF_typeHF_8	0.002	0.002	0.001	0.002
actvtys:HF_HF_2 actvttyw:HF_HF_2 actvtys:HF_HF_3				
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				

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session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2 0.410
activitystanding:HF_typeHF_3 0.506 0.208
activitywalking:HF_typeHF_3 0.209 0.503 0.413
activitystanding:HF_typeHF_5 0.589 0.243 0.764
activitywalking:HF_typeHF_5 0.244 0.586 0.316
activitystanding:HF_typeHF_7 0.459 0.189 0.595
activitywalking:HF_typeHF_7 0.190 0.456 0.246
activitystanding:HF_typeHF_8 0.450 0.185 0.584
activitywalking:HF_typeHF_8 0.186 0.447 0.241
sexM:HF_typeHF_2 0.000 0.001 0.000
sexM:HF_typeHF_3 0.000 0.001 0.000
sexM:HF_typeHF_5 0.000 0.001 0.000
sexM:HF_typeHF_7 0.000 0.001 0.000
sexM:HF_typeHF_8 0.000 0.001 0.000
actvtyw:HF_HF_3 actvtys:HF_HF_5 actvtyw:HF_HF_5

day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7

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session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3
activitywalking:HF_typeHF_3
activitystanding:HF_typeHF_5 0.315
activitywalking:HF_typeHF_5 0.762          0.414
activitystanding:HF_typeHF_7 0.246          0.693          0.287
activitywalking:HF_typeHF_7 0.592          0.286          0.690
activitystanding:HF_typeHF_8 0.241          0.679          0.281
activitywalking:HF_typeHF_8 0.582          0.281          0.678
sexM:HF_typeHF_2          0.001          0.000          0.002
sexM:HF_typeHF_3          0.002          0.000          0.002
sexM:HF_typeHF_5          0.002          0.000          0.002
sexM:HF_typeHF_7          0.001          -0.001          0.002
sexM:HF_typeHF_8          0.001          0.000          0.002
                                actvtys:HF_HF_7 actvtyw:HF_HF_7 actvtys:HF_HF_8
day2
session2
session3
session4
activitystanding
activitywalking
sexM
HF_typeHF_2
HF_typeHF_3
HF_typeHF_5
HF_typeHF_7
HF_typeHF_8
day2:HF_typeHF_2
day2:HF_typeHF_3
day2:HF_typeHF_5
day2:HF_typeHF_7
day2:HF_typeHF_8
session2:HF_typeHF_2
session3:HF_typeHF_2
session4:HF_typeHF_2
session2:HF_typeHF_3
session3:HF_typeHF_3
session4:HF_typeHF_3
session2:HF_typeHF_5
session3:HF_typeHF_5
session4:HF_typeHF_5
session2:HF_typeHF_7
session3:HF_typeHF_7
session4:HF_typeHF_7
session2:HF_typeHF_8
session3:HF_typeHF_8
session4:HF_typeHF_8
activitystanding:HF_typeHF_2
activitywalking:HF_typeHF_2
activitystanding:HF_typeHF_3

```

activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7	0.412			
activitystanding:HF_typeHF_8	0.530	0.219		
activitywalking:HF_typeHF_8	0.219	0.527	0.412	
sexM:HF_typeHF_2	0.000	0.001	0.000	
sexM:HF_typeHF_3	0.000	0.001	0.000	
sexM:HF_typeHF_5	0.000	0.001	0.000	
sexM:HF_typeHF_7	0.000	0.001	0.000	
sexM:HF_typeHF_8	0.000	0.001	0.001	
	actvtyw:HF_HF_8	sM:HF_HF_2	sM:HF_HF_3	sM:HF_HF_5
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				

activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2	0.001			
sexM:HF_typeHF_3	0.001	0.496		
sexM:HF_typeHF_5	0.001	0.516	0.566	
sexM:HF_typeHF_7	0.001	0.481	0.527	0.549
sexM:HF_typeHF_8	0.002	0.479	0.524	0.546
	sM:HF_HF_7			
day2				
session2				
session3				
session4				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
session2:HF_typeHF_2				
session3:HF_typeHF_2				
session4:HF_typeHF_2				
session2:HF_typeHF_3				
session3:HF_typeHF_3				
session4:HF_typeHF_3				
session2:HF_typeHF_5				
session3:HF_typeHF_5				
session4:HF_typeHF_5				
session2:HF_typeHF_7				
session3:HF_typeHF_7				
session4:HF_typeHF_7				
session2:HF_typeHF_8				
session3:HF_typeHF_8				
session4:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				
activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2				
sexM:HF_typeHF_3				
sexM:HF_typeHF_5				
sexM:HF_typeHF_7				
sexM:HF_typeHF_8	0.509			

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-5.6179143	-0.4511729	0.1113639	0.5790220	3.3445918

Number of Observations: 2638

Number of Groups:

person	HF_type	%in%	person
24			144

```
> anova(m3_arh1, type = "marginal")
```

	numDF	denDF	F-value	p-value
(Intercept)	1	2458	4070.811	<.0001
day	1	2458	6.674	0.0098
session	3	2458	7.595	<.0001
activity	2	2458	21.680	<.0001
sex	1	22	0.001	0.9699
HF_type	5	110	26.333	<.0001
day:HF_type	5	2458	15.630	<.0001
session:HF_type	15	2458	2.374	0.0021
activity:HF_type	10	2458	46.125	<.0001
sex:HF_type	5	110	0.796	0.5549

```
> prune_m3_arh1_ML <- lme(
+   fixed = log(HF_value) ~ (day + session + activity) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
> m3_arh1_ML <- lme(
+   fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
> anova(prune_m3_arh1_ML, m3_arh1_ML)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
prune_m3_arh1_ML	1	53	-1199.761	-888.2392	652.8807			
m3_arh1_ML	2	59	-1193.509	-846.7205	655.7547	1 vs 2	5.747956	0.452

Examining the coefficients, we saw that we should remove the variables Sex and the interaction term of Sensor

and Sex, as the model without those terms has better AIC and BIC, and the p-value from the likelihood ratio test is not statistically significant (>0.4). We also dropped the interaction term of Sensor and Session and the variable Session; although the p-value from the likelihood ratio test of dropping them is 0.0031, which is fairly small and suggests including them, dropping them reduces the degrees of freedom from 53 to 35, decreases the BIC by ~ 100 (-888 to -991) and only increases the AIC by 3 (-1197 to -1200). Table 5 shows the likelihood ratio test between the final model (dropping both Sex and Session variables) and the previous, unpruned model.

```
> final_model_ML <- lme(
+   fixed = log(HF_value) ~ (day + activity) * HF_type + session,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
> prune_m3_no_session_at_all <- lme(
+   fixed = log(HF_value) ~ (day + activity) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
>
> anova(final_model_ML, prune_m3_arh1_ML)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
final_model_ML	1	38	-1194.101	-970.7460	635.0508			
prune_m3_arh1_ML	2	53	-1199.761	-888.2392	652.8807	1 vs 2	35.65983	0.002

```
> anova(final_model_ML, prune_m3_no_session_at_all)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
final_model_ML	1	38	-1194.101	-970.7460	635.0508			
prune_m3_no_session_at_all	2	35	-1197.075	-991.3524	633.5373	1 vs 2		
final_model_ML								
prune_m3_no_session_at_all								

```
> anova(prune_m3_arh1_ML, prune_m3_no_session_at_all)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
prune_m3_arh1_ML	1	53	-1199.761	-888.2392	652.8807			
prune_m3_no_session_at_all	2	35	-1197.075	-991.3524	633.5373	1 vs 2		

```
prune_m3_arh1_ML
```

```
prune_m3_no_session_at_all 38.68676 0.0031
```

```
> # The result shows shouldn't include /session
> final1_model_ML <- lme(
+   fixed = log(HF_value) ~ (day + activity) * HF_type + session,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
>
> final2_model_ML <- lme(
+   fixed = log(HF_value) ~ (day + activity) * HF_type + session,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day/session),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
> anova(final1_model_ML, final2_model_ML)
```

	Model	df	AIC	BIC	logLik
final1_model_ML	1	38	-1194.101	-970.7460	635.0508
final2_model_ML	2	38	-1003.676	-780.3203	539.8379

```
> mdl8_arh1_ml <- lme(
+   fixed = log(HF_value) ~ (day + session + activity + sex) * HF_type,
+   random = list(
+     person = ~ 1 + day,
+     HF_type = ~ 1
+   ),
+   correlation = corAR1(form = ~ time_step | person/HF_type/day),
+   weights = varIdent(form = ~ 1 | HF_type),
+   data = data_clean,
+   method = "ML",
+   control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
>
> fin_mdl_ml <- lme(
+   fixed = log(HF_value) ~ (day + activity) * HF_type,
+   random = list(
+     person = ~ 1 + day,
```

```

+   HF_type = ~ 1
+ ),
+ correlation = corAR1(form = ~ time_step | person/HF_type/day),
+ weights = varIdent(form = ~ 1 | HF_type),
+ data = data_clean,
+ method = "ML",
+ control = lmeControl(msMaxIter = 200) #basically, increase iterations
+ )
>
>
> anova_fin <- anova.mdl8_arh1_ml, fin.mdl_ml)
> print(anova_fin)

```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
mdl8_arh1_ml	1	59	-1193.509	-846.7205	655.7547			
fin.mdl_ml	2	35	-1197.075	-991.3524	633.5373	1 vs 2	44.43472	0.0068

```

> anova_fin_clean <- data.frame(
+   Model = c("Model with Session and Sex Variables", "Final Model, Drops Session and Sex Variables"),
+   Df = as.numeric(anova_fin$df),
+   AIC = round(as.numeric(anova_fin$AIC)),
+   BIC = round(as.numeric(anova_fin$BIC)),
+   LogLik = round(as.numeric(anova_fin$logLik)),
+   LRT_Stat = c(NA, round(anova_fin$L.Ratio[2], 2)),
+   p_value = c(NA, anova_fin$p_value[2]))
>
>
> anova_fin_clean <- anova_fin_clean %>%
+   mutate(
+     LRT_Stat = ifelse(is.na(LRT_Stat), "--", as.character(LRT_Stat)),
+     p_value = ifelse(is.na(p_value), "--",
+       # eps defines the threshold for "<"
+       # digits defines the rounding for values above that threshold
+       format.pval(p_value, eps = 0.0001, digits = 4))
+   )
>
> col_labels <- c("Model Version",
+   "\\shortstack{Degrees of\\Freedom}",
+   "AIC",
+   "BIC",
+   "Log-Likelihood",
+   "\\shortstack{Likelihood Ratio\\Test Stat}",
+   "P-Value")
>
> kable(anova_fin_clean,
+   format="latex",
+   col.names = col_labels,
+   align = "lrrrrrr",
+   booktabs = TRUE, #, #right aligns numbers
+   escape = FALSE,
+   caption = "\\footnotesize{Likelihood test between unpruned model and the final model removing t",
+   kable_styling(latex_options = c("scale_down", "hold_position"),
+   full_width = FALSE,
+   table.envir = "table") %>%

```

```

+   add_header_above(c("Likelihood Ratio Test for Final Model Pruning" = 7),
+                       bold = TRUE, align = "l") %>%
+   row_spec(0, bold = TRUE)

```

Table 5: Likelihood test between unpruned model and the final model removing the Sex and Session variables and their respective interaction with session variables. The final model significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using ML, as they have different fixed effects.

Likelihood Ratio Test for Final Model Pruning						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model with Session and Sex Variables	59	-1194	-847	656	—	—
Final Model, Drops Session and Sex Variables	35	-1197	-991	634	44.43	0.006789

```

> # library(dplyr)
> # res <- residuals(final_model_ML_1, type = "normalized") # 2. Add them back to your dataframe for pl
> #
> # # Calculate the variance of residuals for each session
> # var_summary <- data_clean %>%
> #   group_by(session) %>%
> #   summarize(variance = var(res, na.rm = TRUE),
> #             std_dev = sd(res, na.rm = TRUE))
> #
> # print(var_summary)

> # # Use type = "response" to see the un-corrected errors
> # res_raw <- residuals(final_model_ML_1, type = "response")
> #
> # data_clean$res_raw_vals <- res_raw
> #
> # # Now recalculate the variance
> # var_summary_raw <- data_clean %>%
> #   group_by(session) %>%
> #   summarize(variance = var(res_raw_vals, na.rm = TRUE),
> #             std_dev = sd(res_raw_vals, na.rm = TRUE))
> #
> # print(var_summary_raw)

> print(as.data.frame(summary(final_model_REML)$tTable))

```

	Value	Std.Error	DF	t-value
(Intercept)	4.38235800	0.05125173	2476	85.5065424
day2	-0.14731711	0.05715785	2476	-2.5773731
activitystanding	-0.07794965	0.01738539	2476	-4.4836295
activitywalking	0.09343969	0.02119907	2476	4.4077263
HF_typeHF_2	-0.60725986	0.07351840	115	-8.2599709
HF_typeHF_3	-0.50544294	0.06413996	115	-7.8803122
HF_typeHF_5	-0.30916643	0.06013399	115	-5.1412921
HF_typeHF_7	-0.05383023	0.06671293	115	-0.8068935
HF_typeHF_8	-0.73801614	0.06765450	115	-10.9086040
day2:HF_typeHF_2	0.46906513	0.07910936	2476	5.9293250
day2:HF_typeHF_3	0.46733432	0.06133323	2476	7.6195943
day2:HF_typeHF_5	0.41343574	0.05309389	2476	7.7868799
day2:HF_typeHF_7	0.30625195	0.06720837	2476	4.5567530

day2:HF_typeHF_8	0.46598135	0.06840363	2476	6.8122315
activitystanding:HF_typeHF_2	-0.10590601	0.02744309	2476	-3.8591141
activitywalking:HF_typeHF_2	-0.46130890	0.03370466	2476	-13.6867969
activitystanding:HF_typeHF_3	0.17845251	0.02127605	2476	8.3874819
activitywalking:HF_typeHF_3	0.17129203	0.02603197	2476	6.5800647
activitystanding:HF_typeHF_5	0.05153090	0.01839152	2476	2.8018835
activitywalking:HF_typeHF_5	-0.10978767	0.02244613	2476	-4.8911628
activitystanding:HF_typeHF_7	-0.03816032	0.02346988	2476	-1.6259273
activitywalking:HF_typeHF_7	-0.14552856	0.02877332	2476	-5.0577595
activitystanding:HF_typeHF_8	0.16252431	0.02387548	2476	6.8071639
activitywalking:HF_typeHF_8	0.09151240	0.02915125	2476	3.1392271

	p-value
(Intercept)	0.000000e+00
day2	1.001280e-02
activitystanding	7.672771e-06
activitywalking	1.089150e-05
HF_typeHF_2	2.801101e-13
HF_typeHF_3	2.038172e-12
HF_typeHF_5	1.130940e-06
HF_typeHF_7	4.213935e-01
HF_typeHF_8	1.881187e-19
day2:HF_typeHF_2	3.466186e-09
day2:HF_typeHF_3	3.598956e-14
day2:HF_typeHF_5	1.001636e-14
day2:HF_typeHF_7	5.447184e-06
day2:HF_typeHF_8	1.202188e-11
activitystanding:HF_typeHF_2	1.167075e-04
activitywalking:HF_typeHF_2	3.686654e-41
activitystanding:HF_typeHF_3	8.224389e-17
activitywalking:HF_typeHF_3	5.718848e-11
activitystanding:HF_typeHF_5	5.120025e-03
activitywalking:HF_typeHF_5	1.066698e-06
activitystanding:HF_typeHF_7	1.040924e-01
activitywalking:HF_typeHF_7	4.552809e-07
activitystanding:HF_typeHF_8	1.244482e-11
activitywalking:HF_typeHF_8	1.713897e-03

```
> final_model_REML_t_table <- as.data.frame(summary(final_model_REML)$tTable)
> print(final_model_REML_t_table)
```

	Value	Std.Error	DF	t-value
(Intercept)	4.38235800	0.05125173	2476	85.5065424
day2	-0.14731711	0.05715785	2476	-2.5773731
activitystanding	-0.07794965	0.01738539	2476	-4.4836295
activitywalking	0.09343969	0.02119907	2476	4.4077263
HF_typeHF_2	-0.60725986	0.07351840	115	-8.2599709
HF_typeHF_3	-0.50544294	0.06413996	115	-7.8803122
HF_typeHF_5	-0.30916643	0.06013399	115	-5.1412921
HF_typeHF_7	-0.05383023	0.06671293	115	-0.8068935
HF_typeHF_8	-0.73801614	0.06765450	115	-10.9086040
day2:HF_typeHF_2	0.46906513	0.07910936	2476	5.9293250
day2:HF_typeHF_3	0.46733432	0.06133323	2476	7.6195943
day2:HF_typeHF_5	0.41343574	0.05309389	2476	7.7868799
day2:HF_typeHF_7	0.30625195	0.06720837	2476	4.5567530
day2:HF_typeHF_8	0.46598135	0.06840363	2476	6.8122315

activitystanding:HF_typeHF_2	-0.10590601	0.02744309	2476	-3.8591141
activitywalking:HF_typeHF_2	-0.46130890	0.03370466	2476	-13.6867969
activitystanding:HF_typeHF_3	0.17845251	0.02127605	2476	8.3874819
activitywalking:HF_typeHF_3	0.17129203	0.02603197	2476	6.5800647
activitystanding:HF_typeHF_5	0.05153090	0.01839152	2476	2.8018835
activitywalking:HF_typeHF_5	-0.10978767	0.02244613	2476	-4.8911628
activitystanding:HF_typeHF_7	-0.03816032	0.02346988	2476	-1.6259273
activitywalking:HF_typeHF_7	-0.14552856	0.02877332	2476	-5.0577595
activitystanding:HF_typeHF_8	0.16252431	0.02387548	2476	6.8071639
activitywalking:HF_typeHF_8	0.09151240	0.02915125	2476	3.1392271
	p-value			
(Intercept)	0.000000e+00			
day2	1.001280e-02			
activitystanding	7.672771e-06			
activitywalking	1.089150e-05			
HF_typeHF_2	2.801101e-13			
HF_typeHF_3	2.038172e-12			
HF_typeHF_5	1.130940e-06			
HF_typeHF_7	4.213935e-01			
HF_typeHF_8	1.881187e-19			
day2:HF_typeHF_2	3.466186e-09			
day2:HF_typeHF_3	3.598956e-14			
day2:HF_typeHF_5	1.001636e-14			
day2:HF_typeHF_7	5.447184e-06			
day2:HF_typeHF_8	1.202188e-11			
activitystanding:HF_typeHF_2	1.167075e-04			
activitywalking:HF_typeHF_2	3.686654e-41			
activitystanding:HF_typeHF_3	8.224389e-17			
activitywalking:HF_typeHF_3	5.718848e-11			
activitystanding:HF_typeHF_5	5.120025e-03			
activitywalking:HF_typeHF_5	1.066698e-06			
activitystanding:HF_typeHF_7	1.040924e-01			
activitywalking:HF_typeHF_7	4.552809e-07			
activitystanding:HF_typeHF_8	1.244482e-11			
activitywalking:HF_typeHF_8	1.713897e-03			

5.7 Final Model

The final model's formula is as follows (where S means sensor number):

$$\begin{aligned}
 \ln(\widehat{\text{HF}}) = & 4.382 - 0.147(\text{Day}_2) - 0.078(\text{Stand}) + 0.093(\text{Walk}) \\
 & - 0.607(S_2) - 0.505(S_3) - 0.309(S_5) - 0.054(S_7) - 0.738(S_8) \\
 & + 0.469(\text{Day}_2 \cdot S_2) + 0.467(\text{Day}_2 \cdot S_3) + 0.413(\text{Day}_2 \cdot S_5) \\
 & + 0.306(\text{Day}_2 \cdot S_7) + 0.466(\text{Day}_2 \cdot S_8) \\
 & - 0.106(\text{Stand} \cdot S_2) - 0.461(\text{Walk} \cdot S_2) + 0.178(\text{Stand} \cdot S_3) \\
 & + 0.171(\text{Walk} \cdot S_3) + 0.052(\text{Stand} \cdot S_5) - 0.110(\text{Walk} \cdot S_5) \\
 & - 0.038(\text{Stand} \cdot S_7) - 0.146(\text{Walk} \cdot S_7) + 0.163(\text{Stand} \cdot S_8) \\
 & + 0.092(\text{Walk} \cdot S_8)
 \end{aligned}$$

```

> #this keeps failing to format, give up
> # 1. Extract the t-table from the model summary
>

```

```

> # 2. Select specific columns and format
> final_model_REML_t_table_clean <- final_model_REML_t_table %>%
+   rownames_to_column(var = "Term") %>%
+   select(Term, Value, Std.Error, `p-value`) %>%
+   mutate(
+     # Clean the terms: LaTeX interprets underscores as math subscripts
+     Term = gsub("_", " ", Term),
+     Value = round(Value, 4),
+     Std.Error = round(Std.Error, 4),
+     `p-value` = format.pval(`p-value`, eps = 0.0001, digits = 4)
+   )
>
> # 3. Output as a clean table for your PDF
> kable(final_model_REML_t_table_clean,
+   format = "latex",
+   escape = TRUE,
+   booktabs = TRUE,
+   longtable = TRUE, # Recommended for long interaction lists
+   caption = "\\footnotesize{Fixed effects summary for the final Linear Mixed Model. Estimates represent the impact on log transformed heat flow.}"
+   kable_styling(latex_options = c("striped", "hold_position", "scale_down"))

```

Table 6: Fixed effects summary for the final Linear Mixed Model. Estimates represent the impact on log transformed heat flow.

Term	Value	Std.Error	p-value
(Intercept)	4.3824	0.0513	< 1e-04
day2	-0.1473	0.0572	0.0100128
activitystanding	-0.0779	0.0174	< 1e-04
activitywalking	0.0934	0.0212	< 1e-04
HF typeHF 2	-0.6073	0.0735	< 1e-04
HF typeHF 3	-0.5054	0.0641	< 1e-04
HF typeHF 5	-0.3092	0.0601	< 1e-04
HF typeHF 7	-0.0538	0.0667	0.4213935
HF typeHF 8	-0.7380	0.0677	< 1e-04
day2:HF typeHF 2	0.4691	0.0791	< 1e-04
day2:HF typeHF 3	0.4673	0.0613	< 1e-04
day2:HF typeHF 5	0.4134	0.0531	< 1e-04
day2:HF typeHF 7	0.3063	0.0672	< 1e-04
day2:HF typeHF 8	0.4660	0.0684	< 1e-04
activitystanding:HF typeHF 2	-0.1059	0.0274	0.0001167
activitywalking:HF typeHF 2	-0.4613	0.0337	< 1e-04
activitystanding:HF typeHF 3	0.1785	0.0213	< 1e-04
activitywalking:HF typeHF 3	0.1713	0.0260	< 1e-04
activitystanding:HF typeHF 5	0.0515	0.0184	0.0051200
activitywalking:HF typeHF 5	-0.1098	0.0224	< 1e-04
activitystanding:HF typeHF 7	-0.0382	0.0235	0.1040924
activitywalking:HF typeHF 7	-0.1455	0.0288	< 1e-04
activitystanding:HF typeHF 8	0.1625	0.0239	< 1e-04
activitywalking:HF typeHF 8	0.0915	0.0292	0.0017139

```

> # the variances (tau^2, sigma^2, d00, d11)
> VarCorr(final_model_REML)

          Variance      StdDev      Corr
person =  pdLogChol(1 + day)
(Intercept) 0.006643085      0.08150513 (Intr)
day2         0.019320056      0.13899660 -0.901
HF_type =  pdLogChol(1)
(Intercept) 0.025068674      0.15833090
Residual    0.081666941      0.28577428

> var_person = 0.00664308
> var_day = 0.019320056
> var_sensor = 0.025068674
> var_res = 0.081666941
>
> icc_person = var_person/(var_day+var_sensor+var_res)
> icc_day = var_day/(var_day+var_sensor+var_res)
>
> icc_person

[1] 0.05269957
> icc_day

[1] 0.1532661

```

6 Discussion

The ultimate objective of this study was to model the complex thermoregulatory mechanisms of the human body to assist the EPFL ICE lab in creating personalized thermal comfort profiles for energy-efficient building design. By implementing a Linear Mixed Model, we partitioned the variance in skin heat flow, revealing highly significant inter-individual differences, intra-individual temporal dynamics, and covariance structure in sensors.

6.1 Modeling Multivariate Sensor Data via a Univariate Framework

The experimental context was inherently multivariate, as each participant generated simultaneous measurements across multiple body locations (sensors). To construct a valid covariance structure between these sensors using a univariate model, the dataset was structured in long format, introducing the specific sensor (HF_type) as both a fixed covariate and a nested hierarchical grouping factor. This formulation allowed the model to estimate both marginal and joint measurements. The fixed-effect interactions (e.g., day:HF_type) established the marginal population-average behavior for each specific sensor, while the residual covariance matrix ($\mathbf{R}$) and random-effects matrix ($\mathbf{G}$) successfully captured the joint distributions, mathematically linking the synchronous fluctuations of different sensors on the same person.

6.2 Inter-Individual Variability and Temperature Adaptation

Inter-individual variability (differences between participants experiencing the exact same conditions) was captured by the subject-level random intercept ($d_{00} \approx 0.0066$), which proved that individuals possess unique, unobserved baseline heat flow even when holding all environmental factors constant.

The temperature effect (the transition from 24°C on Day 1 to 18°C on Day 2) was not uniform across the population. The inclusion of a random slope for day ($d_{11} \approx 0.019$) was highly significant, proving that the difference between the same activities on different days (temperature) depends heavily on the specific person. Furthermore, the model estimated a strong negative correlation ($r = -0.9$) between a participant's baseline

intercept and their day slope. Physiologically, this indicates that individuals who naturally radiate more heat at comfortable baseline temperatures (24°C) experience more drastic reductions in heat flow when exposed to cold environments (18°C).

6.3 Intra-Individual Variability and the Effect of Time

Fluctuations in heat flow within the exact same person over the course of the day (intra-individual variability) were driven by two temporal effects: the progression of the scheduled experimental sessions, and the localized carry-over of physiological states between consecutive activities.

6.4 Conclusion